Challenging the Status Quo: Re-evaluating Prophylaxis in VWD

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UNDER THE SPOTLIGHT

Shifting Prophylaxis Paradigms in VWD

The WIL-31 Study in Focus

AGENDA & SPEAKERS

Insights from the WIL-31 Study

Child Living with VWD

Robert F. Sidonio Jr., US

Adult Living with Frequent Nose Bleeds

Ana Boban, HR

Female Experiencing Heavy Menstrual Bleeding

Csongor Kiss, HU

Q&A

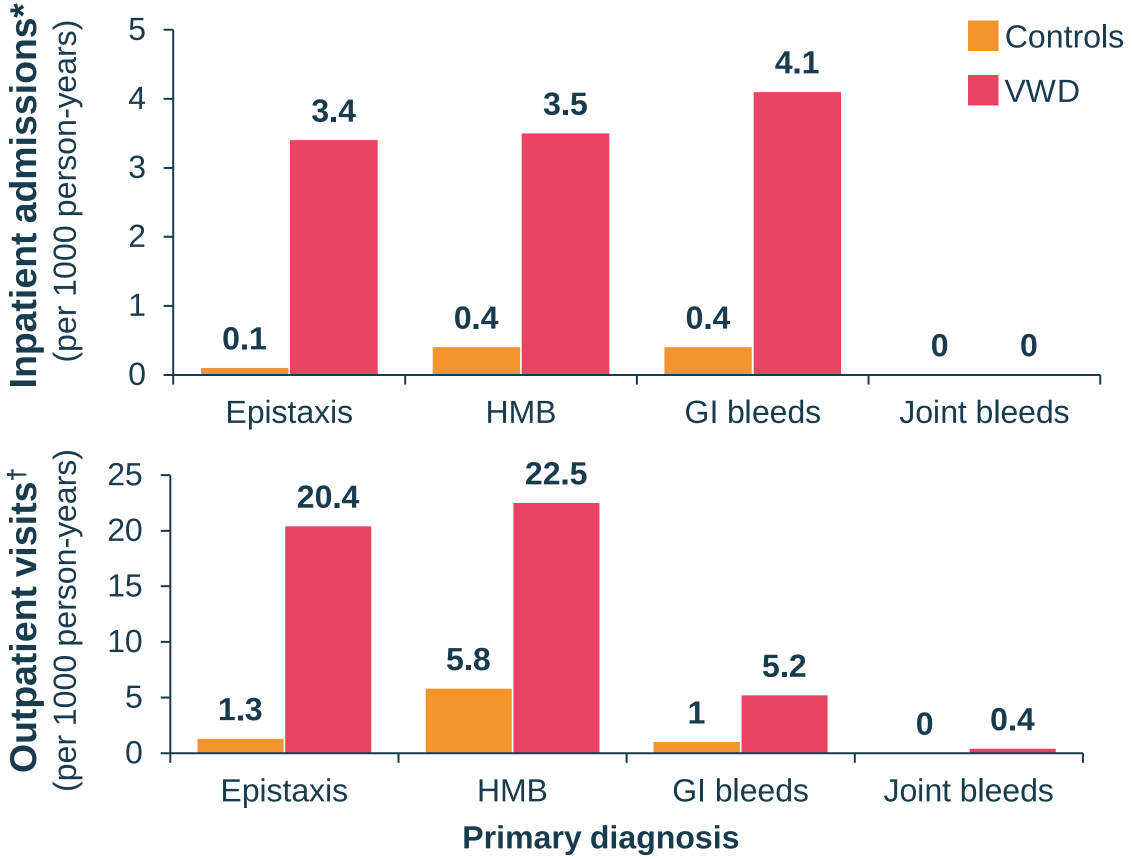
Speakers and Audience

von Willebrand disease (VWD) is the most common inherited bleeding disorder

Up to **1%** **of the world's population** are estimated to have VWD[1]

Up to **90%** of people with VWD are **undiagnosed**[2]

People with VWD have a **higher consumption of healthcare resources** compared to controls[3]



* data from 1987-2009; 56,553 VWD person-years; 267,377 control person-years.
† data from 2004-2009; 14,881 VWD person-years; control 70,034 person years.
HMB: heavy menstrual bleeding; GI: gastrointestinal; VWD: von Willebrand disease.

1. Bowman ML and James PD. *Int J Lab Hematol* 2017; 39(Suppl. 1):61-8; 2. World Federation of Hemophilia. https://www.wfh.org/en/our-work-global/vwd-initiative-program (accessed Jan 2024); 3. Holm E et al. *Haemophilia* 2018; 24:628-33.

Rationale for prophylaxis in VWD

Bleeding can **jeopardise QoL** or be **life-threatening**[1]

Prophylaxis with VWF concentrates can **reduce bleeding rates**[2]

Prophylaxis may **reduce the burden** of VWD and improve **QoL**[2]

Prophylaxis may **reduce HMB** and prevent excessive bleeding during pregnancy[2]

Successful implementation of prophylaxis in haemophilia provides a rationale for its use in VWD[1]

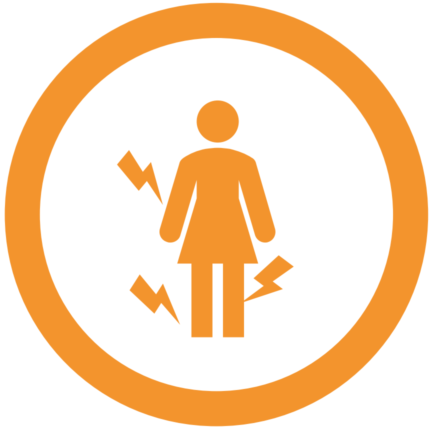
VWF: von Willebrand factor; QoL: quality of life.

1. Holm E et al. *Blood Coagul Fibrinolysis* 2015; 26:383-8; 2. Miesbach W and Berntorp E. *Thromb Res* 2021; 199:67-74.

Joint bleeding is the most common reason for initiating VWF prophylaxis[1]

~50% of patients with type 3 VWD and up to 10% of those with type 1 and type 2 VWD experience joint bleeds[1]

Joint bleeds in VWD are associated with arthropathy and lower QoL,[2,3] similar to patients with haemophilia



Prophylaxis with VWF concentrates reduces joint bleeding for patients with VWD[4]

1. Miesbach W and Berntorp E *Thromb Res* 2021; 199:67-74; 2. van Galen, KPM et al. *Haematologica* 2017; 102: 1486–93;
3. van Galen KP et al. *Haemophilia* 2015; 21:e185–e192; 4. Holm E et al. *Blood Coagul Fibrinolysis* 2015; 26:383-8

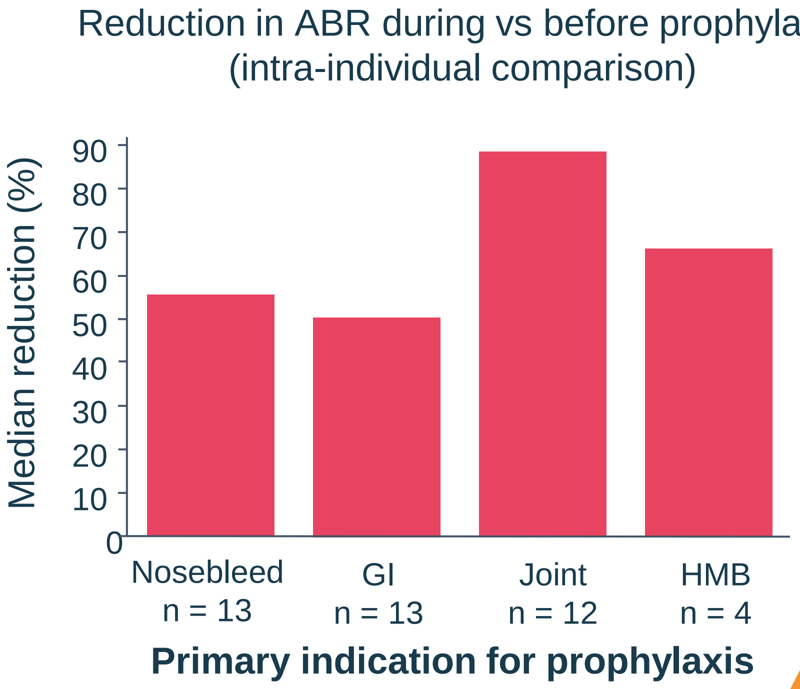
Studies demonstrate a reduction in bleeding in patients with severe VWD on prophylaxis

Mean ABR* was reduced during rVWF prophylaxis vs prior on-demand treatment in adult type 3 VWD (n = 10)[1]

Spontaneous bleeds:	Treated spontaneous bleeds:	Spontaneous or traumatic joint bleeds:
↓48.9%	↓91.6%	↓85.3%
(ABR: 2.70 vs 5.29; 95% CI: -78.4–20.7%)	(ABR: 0.68 vs 8.09; 95% CI: -98.4–-55.4%)	(ABR: 0.34 vs 2.30; 95% CI: -97.2–-23.9%)

Adult type 3 VWD patients switching from pdVWF to rVWF prophylaxis (n = 8) experienced reduced ABRs for treated bleeds **but increased ABRs for all bleeds**[1]

Retrospective, multicentre study of VWF prophylaxis (N = 55)†[2]



* Based on negative binomial regression model; † Patients must have been on a prophylactic regimen at least 6 months or have a history of prophylaxis use for a period of at least 6 months that was subsequently discontinued because it was no longer required.

ABR: annualised bleeding rate; CI: confidence interval; pdVWF: plasma derived VWF; RCT: randomised control trial; rVWF: recombinant VWF.

1. Leebeek FWG et al. *Eur J Haematol* 2023; 111:29-40; 2. Abshire T et al. *Haemophilia* 2013; 19:76-81.

Guidelines recommend prophylaxis in VWD patients with severe symptoms

CLINICAL GUIDELINES

blood advances

ASH ISTH NHF WFH 2021 guidelines on the management of von Willebrand disease

Nathan T. Connell,[1,*] Veronica H. Flood,[2,*] Romina Brignardello-Petersen,[3] Rezan Abdul-Kadir,[4] Alice Arapshian,[5] Susie Couper,[6] Jean M. Grow,[7] Peter Kouides,[8] Michael Laffan,[9] Michelle Lavin,[10] Frank W. G. Leebeek,[11] Sarah H. O'Brien,[12] Margareth C. Ozelo,[13] Alberto Tosetto,[14] Angela C. Weyand,[15] Paula D. James,[16] Mohamad A. Kalot,[17] Nedaa Husainat,[17] and Reem A. Mustafa[17]

*"Patients with a **severe bleeding phenotype (e.g., severe type 1, type 2, or type 3 VWD) may require prophylaxis with VWF concentrate** to prevent bleeding while on antiplatelet or anticoagulant therapy"*

*"In patients with **VWD with a history of severe and frequent bleeds, the guideline panel suggests using long-term prophylaxis** rather than no prophylaxis"*

Proposed definition of prophylaxis:

A period of at least 6 months of treatment with VWF concentrate administered at least once weekly

Connell NT et al. *Blood Adv* 2021; 5:301-25.

Prophylaxis remains underused in VWD

Factors affecting implementation of guidelines:

VWD pathophysiology is complex and involves deficiency of both VWF and FVIII; patients **present with a broad range of bleeding tendencies** ranging from mild to extremely severe[1]

There is **no widely recognised definition for VWF prophylaxis in VWD**[2]

Insufficient prospective data at present that all patients should receive prophylaxis

Prophylaxis recommendations **depend on symptoms, not laboratory findings**

<10%

of patients with severe VWD **receive prophylaxis** with **VWF concentrates**[1]

1. Miesbach W and Berntorp E. *Thromb Res* 2021; 199:67-74; 2. Connell NT et al. *Blood Adv* 2021; 5:301-25.

Challenges in recognising VWD can limit access to appropriate care

Complex diagnosis

Phenotypes are **heterogeneous**, **multiple diagnostic assays** are required, and **genetic analysis is complex**[1-4]

Variable VWF levels

Stress, exercise, hormones and **blood group** can impact VWF levels[1]

Barriers to treatment

Women with VWD experience, on average, a **16-year delay** between symptom onset and diagnosis[5]

1. Lillicrap D and James P. *World Federation of Hemophilia Treatment of Hemophilia Monograph* 2009; No. 47. http://www1.wfh.org/publication/files/pdf-1204.pdf (accessed Jan 2024); 2. Schinco P et al. *Blood Transfus* 2018; 16:371-81; 3. Roberts JC and Flood VH. *Int J Lab Hematol* 2015; 37(Suppl. 1):11-7; 4. Lavin M et al. *Blood* 2017; 130:2344-53; 5. Weyand AC and James PD. *Res Pract Thromb Haemost* 2020; 5:51-4.

Key considerations on the use of VWF prophylaxis in VWD

Patient needs for prophylaxis based on physician surveys[1]

Patients place high value on reducing bleed risk, particularly related to QoL

Importance of **shared decision making** to review risks and benefits

Importance of **available educational material** for clinicians and patients

Other considerations for prophylaxis

Impact of bleeding on QoL, inconvenience of symptoms vs prophylaxis

Treatment with factor requires IV access and often a hospital visit

Cost of prophylaxis, risk/benefit and treatment challenges

IV: intravenous.
Connell NT et al. *Blood Adv* 2021; 5:301-25.

wilate® a human von Willebrand factor/human coagulation factor VIII

Two independent **virus-inactivation** steps[1]

Physiological **1:1 VWF/FVIII ratio** facilitates dosing and monitoring[1-3]

FVIII
VWF

Parallel decay curves of VWF and FVIII may reduce risk of FVIII accumulation and thrombosis[1,2]

No wilate®-related **thrombotic complications** and **no clinically relevant FVIII accumulation** reported in any clinical study[6,8,9]

wilate® was developed **specifically for VWD patients**

1:1

Demonstrated efficacy and **safety** for **surgery, prophylaxis** and **on-demand treatment of bleeds** in patients of all ages[4-7,9]

>18 years of experience in people with **all types of VWD**

FVIII: coagulation factor VIII; VWF: von Willebrand factor.

1. Stadler M et al. *Biologicals* 2006; 34:281-8; 2. Kessler CM et al. *Thromb Haemost* 2011; 106:279-88; 3. Windyga J et al. *Thromb Haemost* 2011; 105:1072-9;
4. Srivastava A et al. *Haemophilia* 2017; 23:264-72; 5. Berntorp E et al. *Haemophilia* 2009; 15:122-30; 6. Sholzberg M et al. *TH Open* 2021, e264-72;
7. wilate® Prescribing Information, revised June 2021; 8. Octapharma AG, Data on file; 9. Sidonio RF et al. *Blood Adv* 2023; accepted.

Summary

- Success of prophylaxis has been demonstrated in haemophilia **providing a rationale for its use in VWD**
- **Prophylaxis is underused in patients with all types of VWD**
- Evidence-based guidelines recommend using **long-term prophylaxis in patients with VWD with a history of severe and frequent bleeds** to reduce bleeding events
- VWD patients receiving prophylaxis have **a lower burden of disease**

UNDER THE SPOTLIGHT

Insights from the WIL-31 Study

Robert F. Sidonio, Jr. MD, MSc

Associate Professor of Pediatrics

Department of Pediatrics,
Emory University School of Medicine,
Atlanta, GA

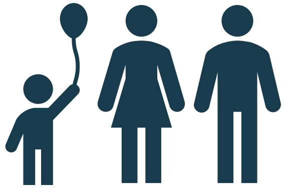
octapharma

WIL-31 is the largest prospective prophylaxis study in VWD with an on-demand run-in study as an intra-individual patient comparator

6 months on-demand with any VWF-containing product

12 months wilate® prophylaxis (2–3 × per week at 20–40 IU/kg)

WIL-29[1] → WIL-31[2]



Eligible for WIL-31 if during WIL-29:

- Experienced ≥ 6 BEs*
- ≥ 2 of these BEs treated with a VWF-containing product

Aim: To investigate the efficacy and safety of wilate® during prophylaxis in previously treated patients with VWD

* Excluding menstrual bleeds.

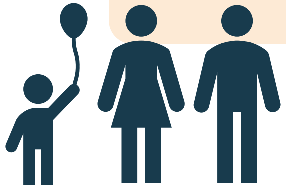
BE: bleeding event; IU: international units; VWD: von Willebrand disease; VWF: von Willebrand factor.

1. https://clinicaltrials.gov/ct2/show/NCT04053699; 2. https://www.clinicaltrials.gov/ct2/show/NCT04052698.

WIL-31: Phase 3, prospective study of wilate® prophylaxis in VWD

Patient criteria

- Male / female ≥ 6 years age
- VWD
  - Type 1 (VWF:RCo < 30 IU/dL)
  - Type 2A, 2B or 2M
  - Type 3
- No history/suspicion of VWF and FVIII inhibitors or history of thromboembolism
- Received on-demand treatment during the WIL-29 study



Endpoints

Primary

- > 50% reduction in total ABR during wilate® prophylaxis compared with prior on-demand treatment

Selected secondary

- Spontaneous ABR
- Consumption of wilate®
- Efficacy of wilate® in the treatment of BEs
- Treatment-emergent AEs

ABR: annualised bleeding rate; AE: adverse event; FVIII: factor VIII; RCo: ristocetin cofactor.
https://clinicaltrials.gov/ct2/show/NCT04052698.

WIL-31: Patient disposition

N = 44
All screened subjects

N = 1
Not treated with wilate®

N = 43
Treated with wilate® (SAF)

N = 10
Excluded due to unconfirmed VWD status

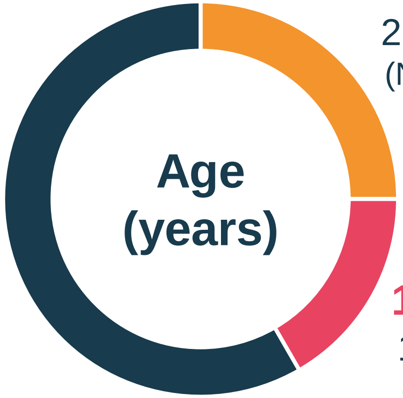
N = 33
Modified full analysis set (mFAS)

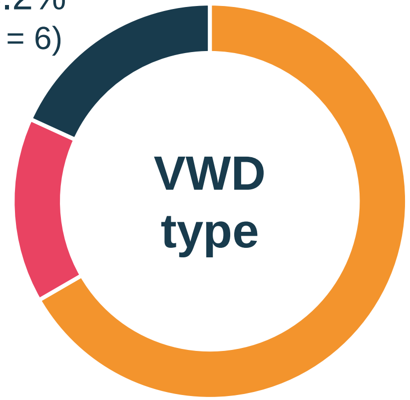
N = 3
Prematurely terminated
2 due to adverse events and
1 with other reasons

N = 30
Completed

SAF: safety set.
Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WIL-31: Demographics





Characteristic median (range) unless otherwise specified	**N = 33**
Age, years	18 (7–61)
Height, cm	167 (118–182)
Weight, kg	61 (21–112)
Race, n (%)	
Black or African American	1 (3.0)
Caucasian	32 (97.0)
Blood type, n (%)	
A	18 (54.5)
B	5 (15.2)
AB	1 (3.0)
O	9 (27.3)
Family history of VWD, n (%)	18 (54.5)

Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WILPROPHY.

WIL-31: wilate prophylaxis reduced total and spontaneous bleeding vs on-demand treatment

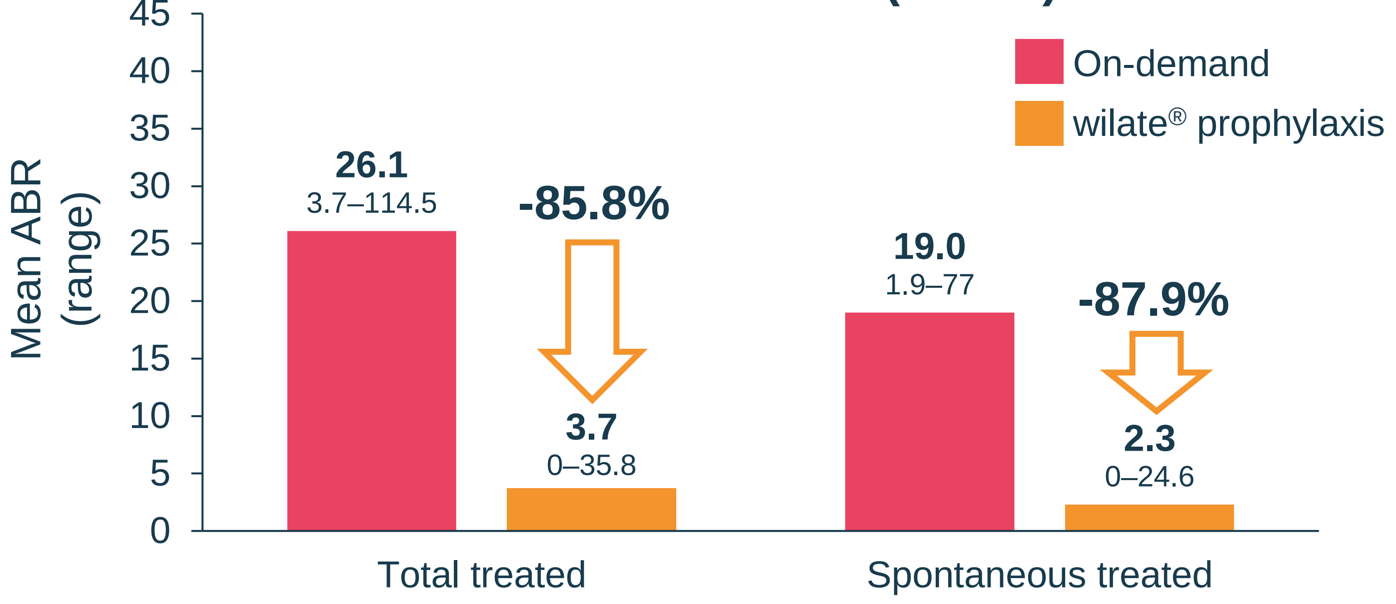
84.4% decrease in mean (SD) total ABR

- On-demand: 33.4 (23.6)
- wilate® prophylaxis: 5.2 (7.7)

Primary endpoint met

≥50% reduction in mean total ABR

ABRs for treated BEs (N = 33)



	Total treated		Spontaneous treated	
	WIL-29	WIL-31	WIL-29	WIL-31
Median (range)	**22.4** (3.7–114.5)	**1** (0–35.8)	**16.4** (1.9–77)	**0** (0–24.6)

SD: standard deviation.

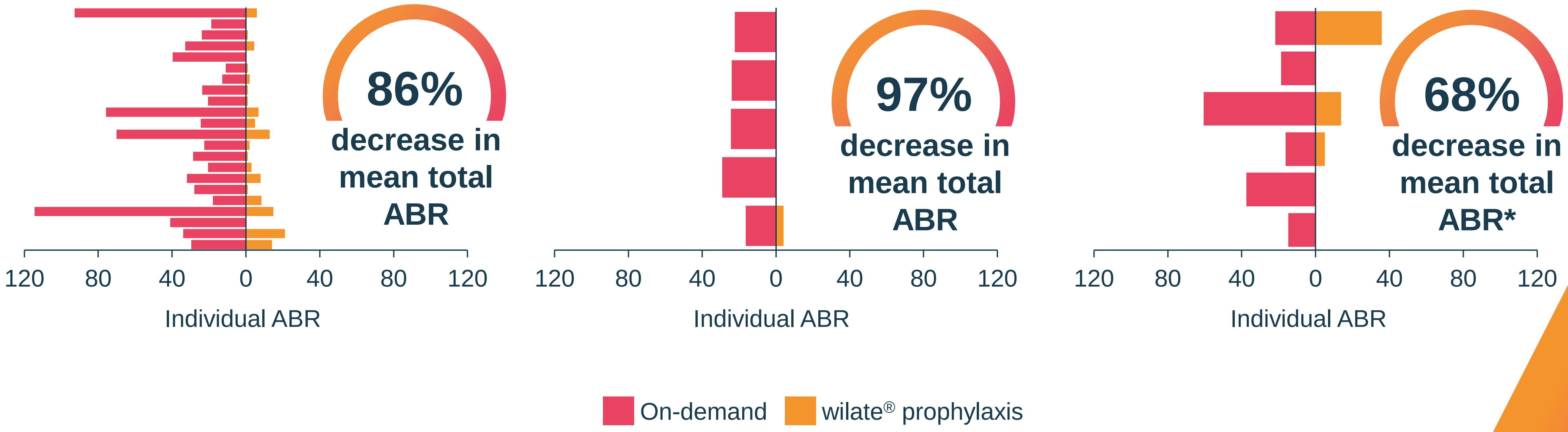
Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WIL-31: Efficacy of wilate® for the prevention of bleeds by VWD type

Type 3 (n = 22)

Type 2A (n = 5)

Severe type 1 (n = 6)



* One patient experienced many bleeds, including 12 GI bleeds during WIL-31 resulting in a higher ABR in WIL-31 compared with WIL-29. This patient is considered an "outlier" among the severe type 1 population and without this patient the reduction is 87%.

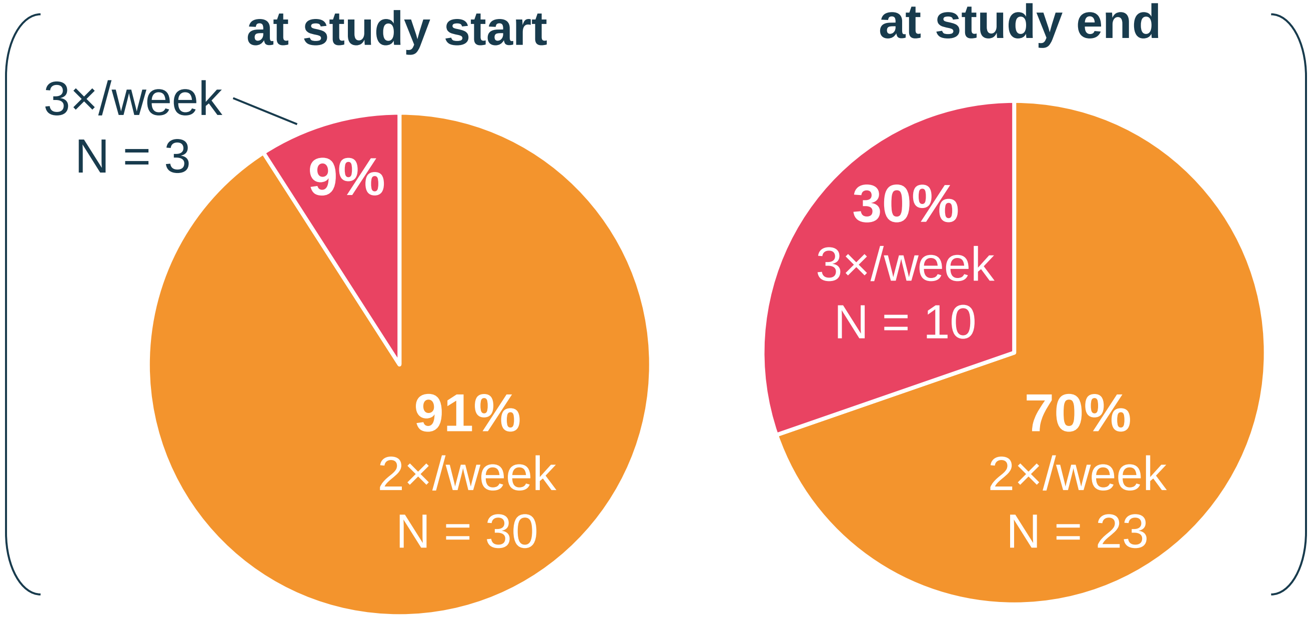
Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WILPROPHY.

WIL-31: wilate® consumption for prophylaxis

Median (range) weekly dose

Prophylactic dosing regimen (N = 33)

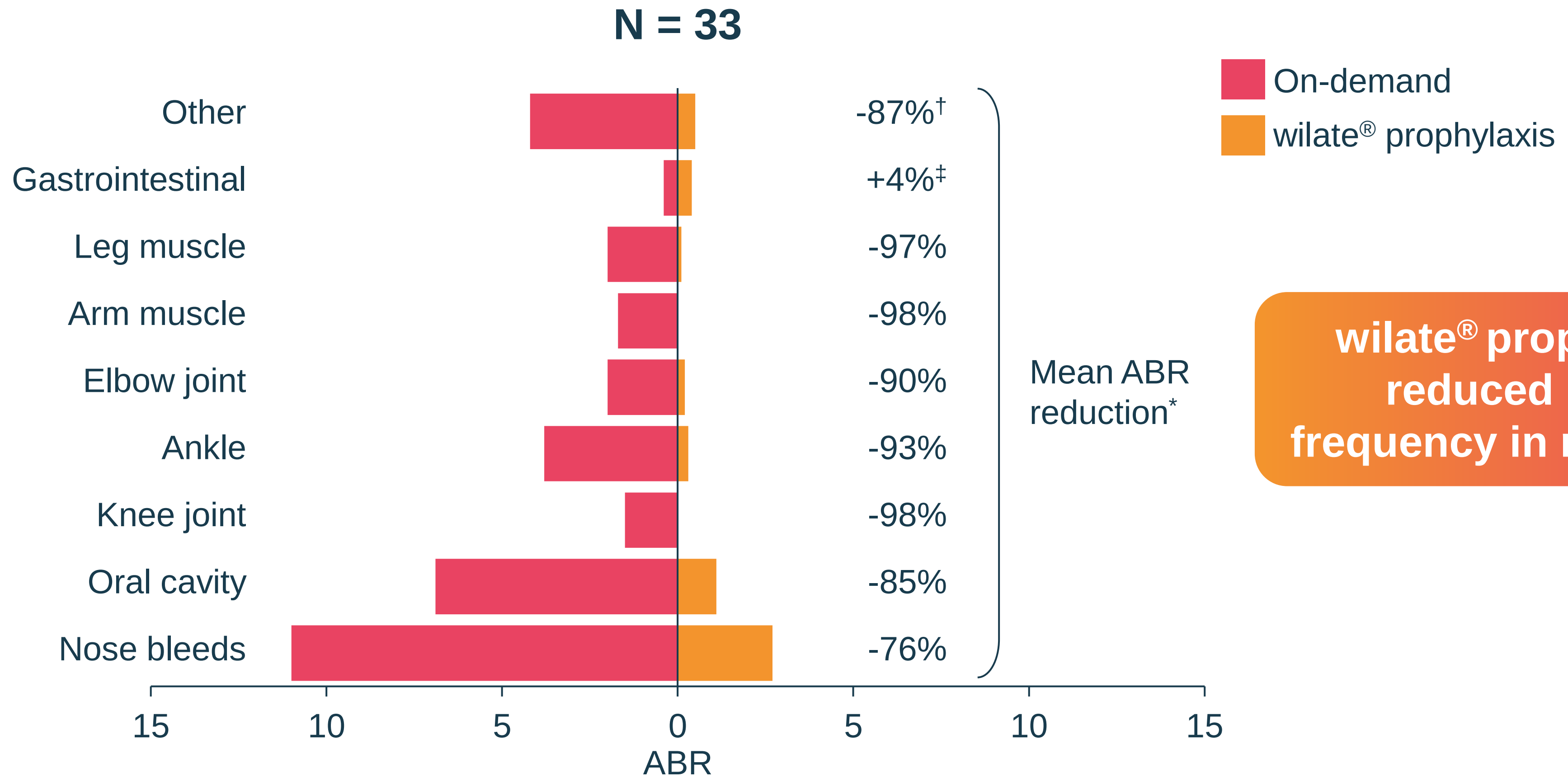


26 patients (79%) required no change in dosing frequency during WIL-31*

All patients were treatment compliant† with their prescribed wilate® prophylaxis regimen

* 7 patients switched from 2× to 3× per week during WIL-31 due to > 2 spontaneous BEs or 1 major spontaneous BE within a month.
† Patients were considered compliant if at least 80% of doses were administered as planned.
Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WIL-31: wilate® prophylaxis reduced bleeding across all sites

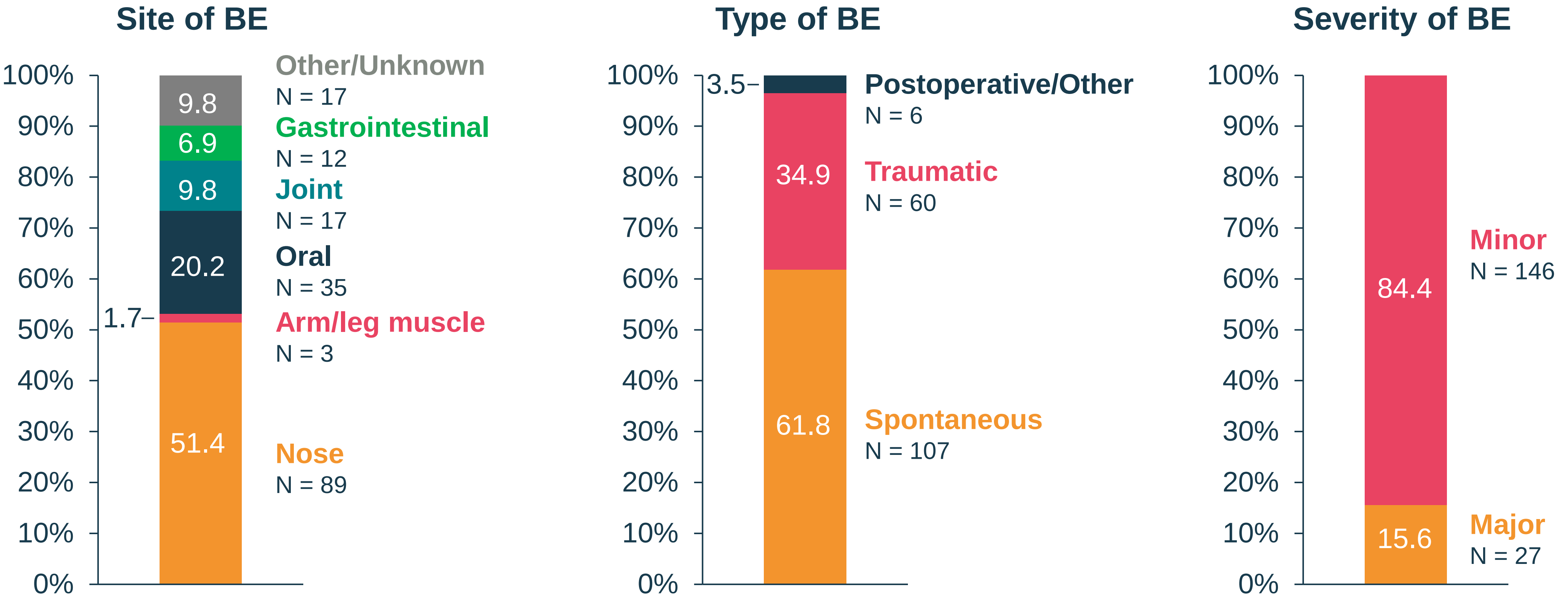


wilate® prophylaxis reduced bleed frequency in most sites

* Reduction in mean ABR (%) = 100-(Mean ABR WIL-31/Mean ABR WIL-29*100). Missing in case ABR = 0 in WIL-29.
† "Other" bleeds in WIL-29 all years include arm, back, buttock, cutaneous, cyst, ecchymosis, haematuria, haemorrhoid, hand, hip, leg, nasal contraction, post-surgery, shoulder, subcutaneous, thigh, toe, tonsil, uterus, and wrist. "Other" bleeds in WIL-31 all years include cutaneous, haemorrhoid, head, hip, ocular, rectal, toe, and tonsil.
‡ One patient experienced many bleeds, including 12 GI bleeds during WIL-31 resulting in a higher ABR in WIL-31 compared with WIL-29. This patient is considered an "outlier" without this patient the reduction is 100%. Without this patient the reduction is 100%.
Octapharma, data on file.

WIL-31: Occurrence of BEs

23/33 patients experienced 173 BEs*



54.5% of patients (18/33) had zero spontaneous treated BEs during WIL-31

* Excludes menstrual bleeds.

Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WIL-31: Haemostatic efficacy of wilate® for treatment of BEs and surgery

22 patients had 121 BEs* treated with wilate®

Median (range) **exposure days per BEs**: 1 (1–5)

Efficacy of treated rated as
“excellent” or **“good”**†

96%
26/27 major

100%
94/94 minor

3 patients underwent 13 surgeries‡

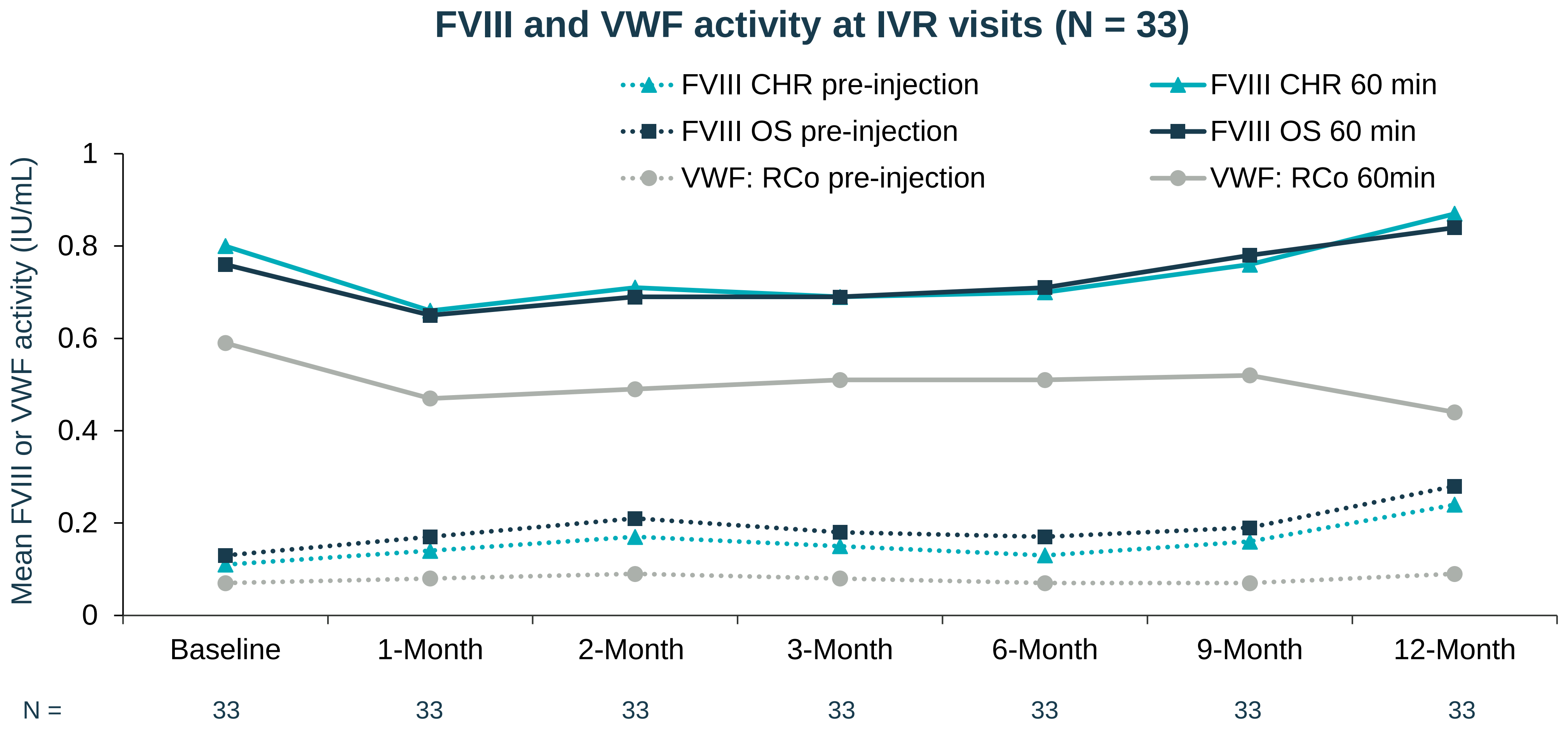
Overall assessment of efficacy for surgery
“excellent” or **“good”**¶

100%
3/3 major

100%
10/10 minor

* Excludes menstrual bleeds. † Excellent: bleeding completely stopped within 3 days for minor bleeds, within 7 days for major bleeds, and within 10 days for gastrointestinal bleeds; good: bleeding completely stopped, but time and/or dose slightly exceeded expectations. ‡ All covered with wilate®. ¶ Overall efficacy using the ‘excellent,’ ‘good,’ moderate,’ and ‘none’ scale took both the intra- and postoperative assessments by the surgeon and haematologist, respectively, into account and was determined by the investigator using a pre-specified algorithm.
Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

WIL-31: No accumulation of FVIII or VWF during wilate® prophylaxis



CHR: chromogenic assay; IVR: In vivo recovery; OS: one-stage assay.
Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742; Octapharma, data on file.

WIL-31: wilate® prophylaxis was well tolerated across all patients

0

Thrombotic events occurred

No patients developed inhibitors during WIL-31

2

Patients had AEs assessed as possibly related to the study treatment by the investigator and led to discontinuation

- 1 patient developed mild chest tightness (3 events)
- 1 patient had hypersensitivity reactions of moderate severity (2 events)

5

Serious AEs occurred in 4 patients,
none considered related to treatment

Sidonio RF et al. *Blood Adv* 2024; doi:10.1182/bloodadvances.2023011742.

Summary

- WIL-31 is the **largest prospective prophylaxis study in VWD** with a prospective on-demand run-in study as an intra-individual comparator
- Primary endpoint of WIL-31 was achieved, with an **84% reduction in total ABR** during wilate® prophylaxis compared with previous on-demand treatment
- wilate® prophylaxis **reduced bleeding across all age groups, all VWD types and all bleed sites** compared with previous on-demand treatment
- **Prophylaxis during WIL-31 was well tolerated**, with no accumulation of VWF or FVIII, thrombotic events or serious treatment related adverse events observed

UNDER THE SPOTLIGHT

Child Living with VWD

octapharma

Ahmed has type 3 VWD and suffered frequent bleeds during on-demand treatment

UNDER THE SPOTLIGHT

Background / 6 months before the WIL-29 study

- Diagnosed with type 3 VWD in 2013 under the age of 1
- No family history of VWD
- 7-year-old male
- Weight 20 kg
- VWF activity (RCo): 1%
- VWF Antigen: 3%
- 37 bleeds treated on-demand with Haemate P® over 6 months
- Two target joints: R ankle, R knee

On-demand treatment during WIL-29

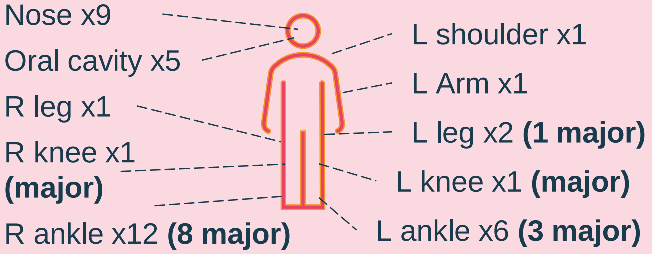
- Treated on-demand (with wilate® and Haemate P®) for 6 months

VWF containing treatment:

- Major bleeds
  - 11/14 treated with wilate®
    - Mean 24.7 IU/kg per infusion
  - 3/14 treated with Haemate P®
- Minor bleeds
  - 23/25 treated with wilate®
    - Mean 27.0 IU/kg per infusion
  - 2/25 treated with Haemate P®

Outcomes during WIL-29

- Total ABR of **76**
- **39** bleeds, of which 14 major
  - 31 spontaneous, 7 traumatic, 1 unknown
- **23% were nose bleeds (all minor)**
  - **Mean duration 10.6 h / nosebleed**
- Two target joints: R ankle, R knee



ABR: Annualised bleeding rate; IU: international units; L: left; R: right; RCo: ristocetin cofactor; VWD: von Willebrand disease; VWF: von Willebrand factor. Octapharma, data on file.

VWD is a serious challenge for paediatric populations

Difficult diagnosis

Paediatric-specific VWD bleeding phenotype such as epistaxis, bruising, and oropharyngeal bleeding is common in children without VWD[1]

Severe bleeding

Following surgery, dental procedures and trauma[1]

Impact on HRQoL

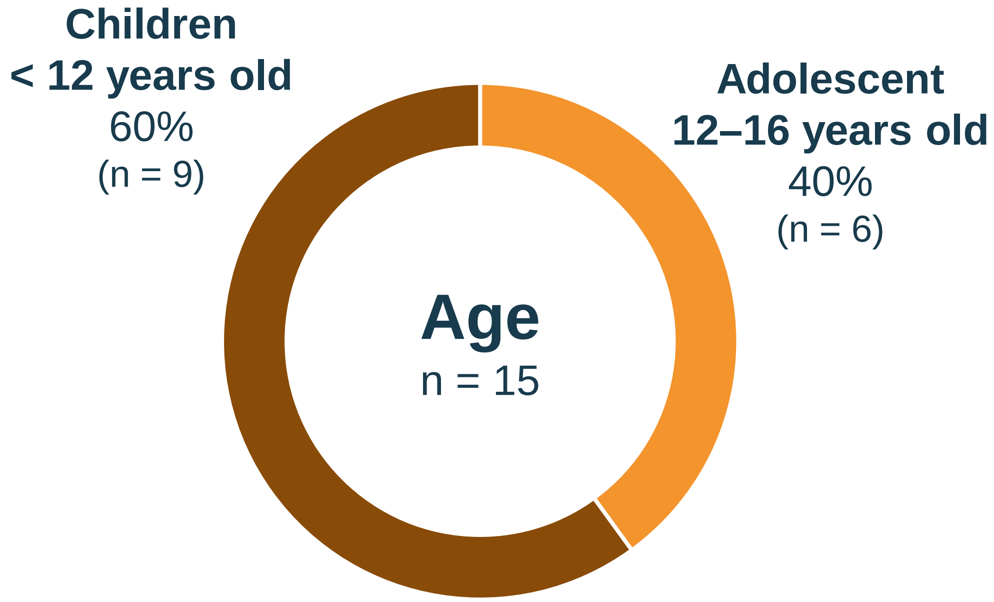
Bleeding symptoms result in decreased HRQoL for children and their caregivers[2,3]

Prophylaxis in children reduces the frequency of recurrent bleeds and may also prevent joint damage[4,5]

HRQoL: Health related quality of life.

1. Sanders YV et al. *Am J Hematol* 2015; 90:1142-8; 2. Olsson A et al. *Haemophilia* 2023; 29:1056-62; 3. de Wee EM et al. *J Thromb Haemost* 2011; 9:502-9;
4. Berntorp and Petrini P. *Blood Coagul Fibrinolysis* 2005; 16 Suppl 1:S23-6; 5. Halimeh S et al. *Thromb Haemost* 2011; 105:597-604.

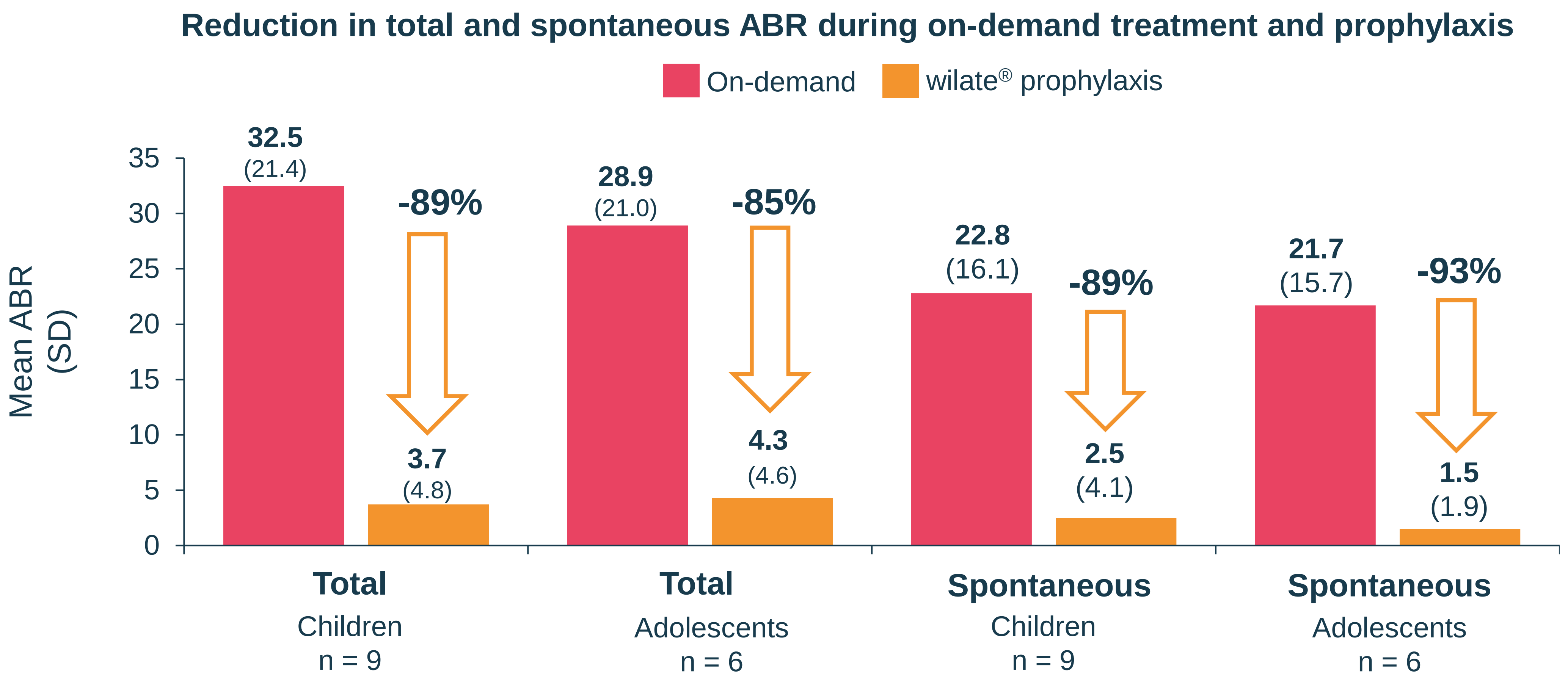
WIL-31: Demographics of children and adolescent population



Characteristic median (range) unless otherwise specified	**Children** 6–11 years old (n = 9)	**Adolescents** 12–16 years old (n = 6)
VWD type, n (%)		
Type 1	2 (22)	1 (17)
Type 2A	2 (22)	1 (17)
Type 3	5 (56)	4 (66)
Age, years	9 (7–11)	15 (12–16)
Height, cm	136 (118–146)	165 (151–182)
Weight, kg	28 (21–41)	67 (40–92)
Caucasian, n (%)	9 (100)	6 (100)
Blood type, n (%)		
A	7 (78)	2 (33)
B	1 (11)	1 (17)
AB	1 (11)	0 (0)
O	0 (0)	3 (50)
Family history of VWD, n (%)	4 (44)	4 (67)

Octapharma, data on file.

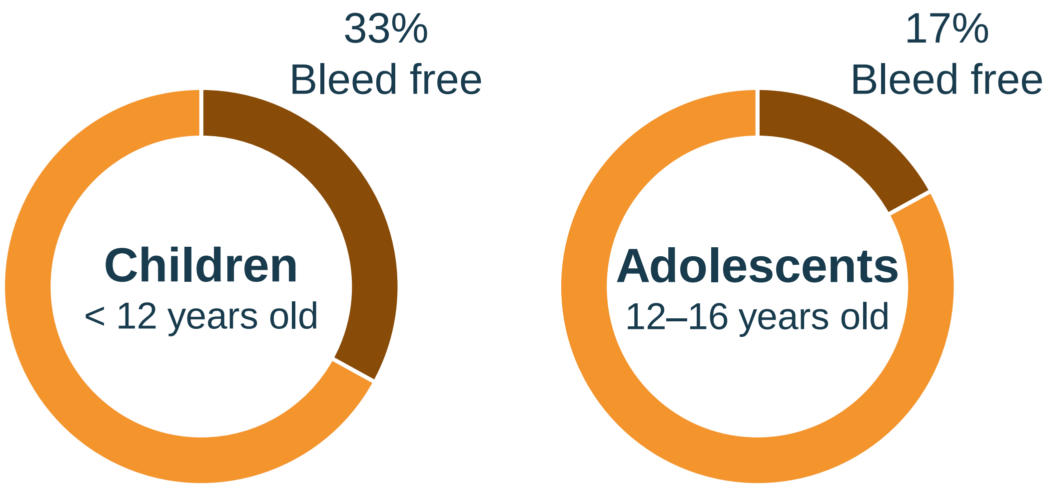
WIL-31: Prophylaxis with wilate® reduced ABRs in children and adolescent patients



SD: standard deviation.
Octapharma, data on file.

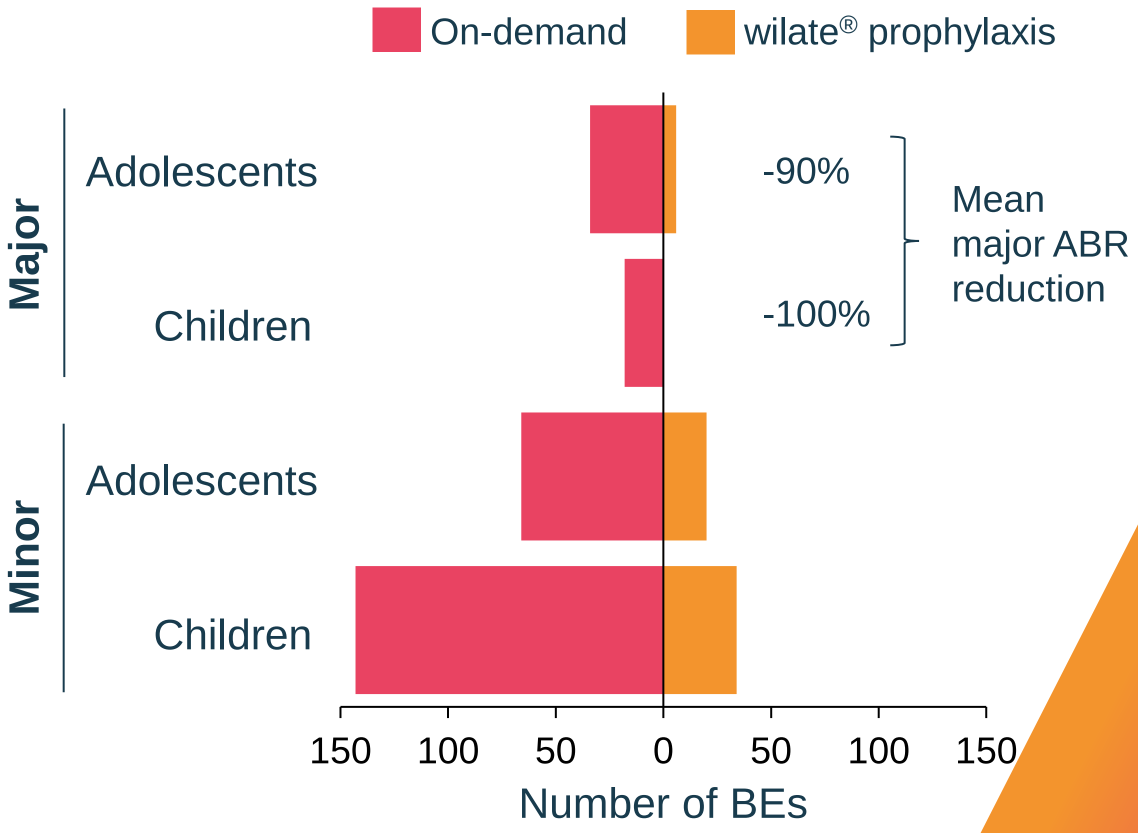
WIL-31: Prophylaxis with wilate® reduced the number and severity of bleeds in children and adolescent patients

Patient bleeding during prophylaxis



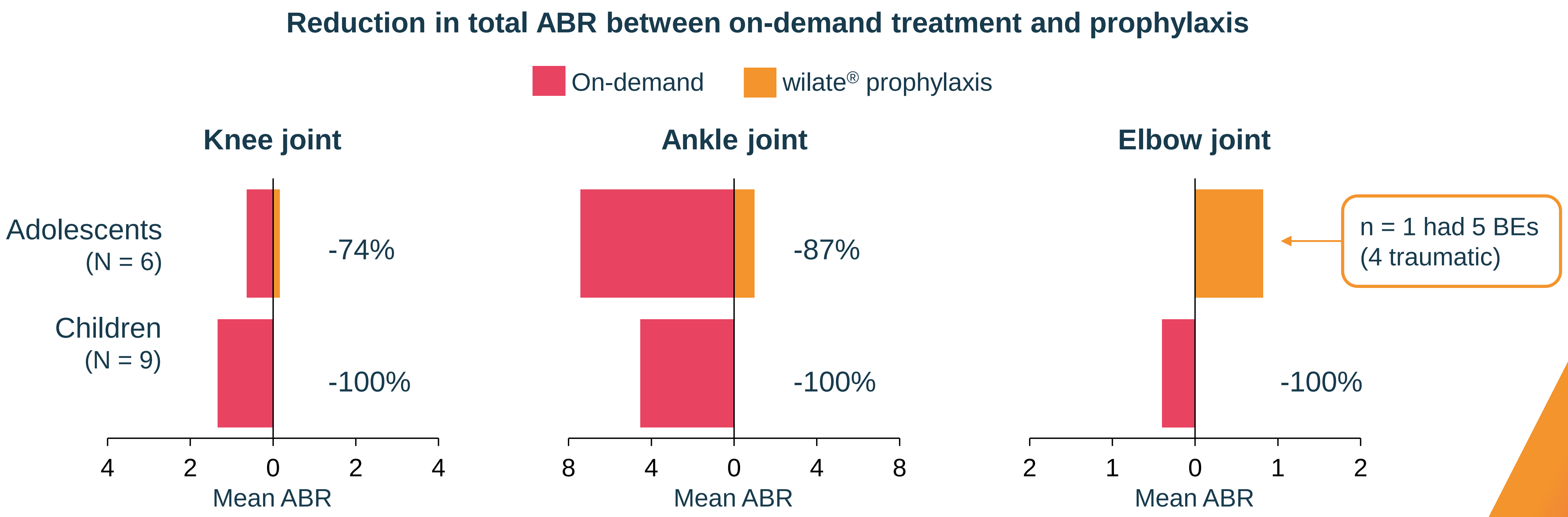
100% of children and adolescents experienced bleeding during on-demand treatment

Bleed severity*



*Excluding menstrual bleeds.
Octapharma, data on file.

WIL-31: wilate® prophylaxis substantially reduced joint bleeding in children and adolescents

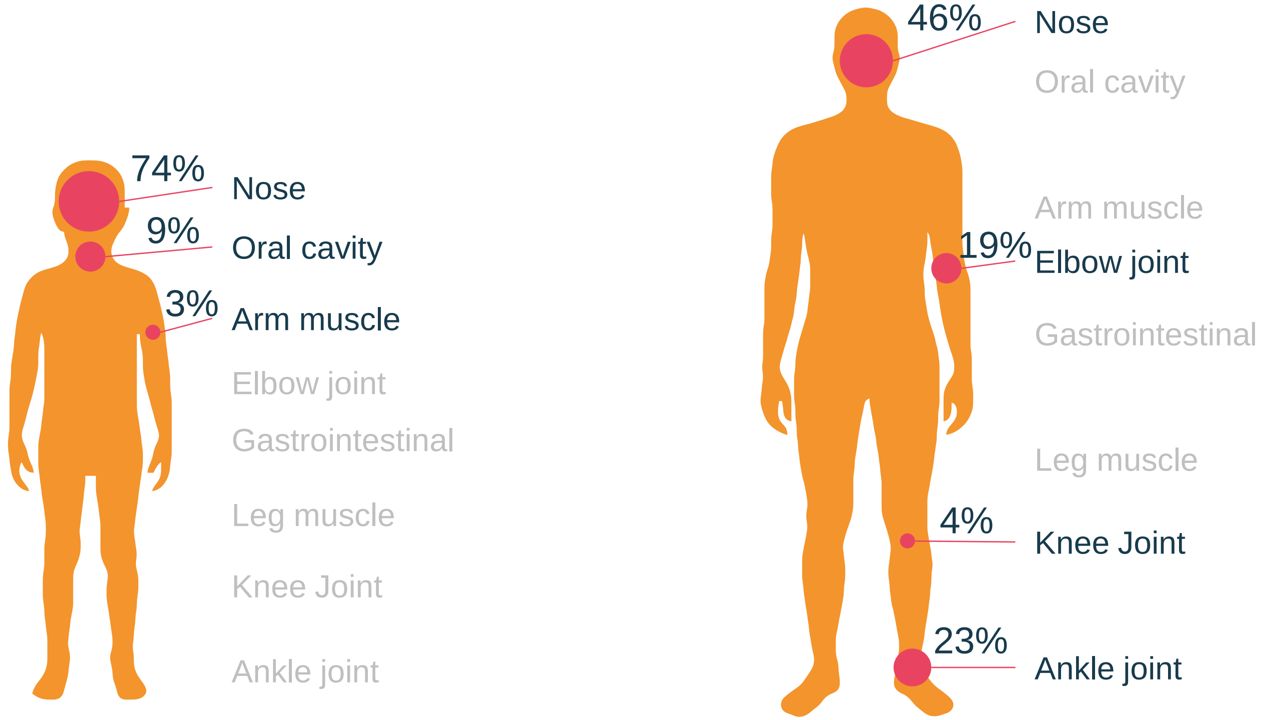


Children experienced no joint bleeds during prophylaxis
Adolescents experienced substantial reductions in knee and ankle joint bleeds

Octapharma, data on file.

WIL-31: wilate® prophylaxis reduced bleeding in children and adolescents

Location of bleeding sites during wilate® prophylaxis



6 - <12 (n=9)
Number of BEs
On-demand: **161**
wilate® prophylaxis: **34***

12 - <17 (n=6)
Number of BEs
On-demand: **100**
wilate® prophylaxis: **26**†

* Other bleeds include cutaneous, head, hip, and toe. † Other bleeds include cutaneous and tonsil.
Octapharma, data on file.

Ahmed experienced a reduction in rate and duration of bleeds during prophylaxis with wilate®

Prophylaxis with wilate® during WIL-31

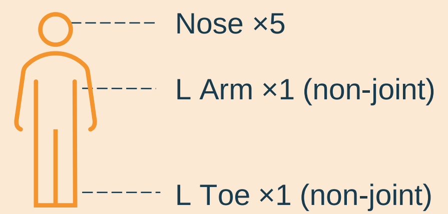
- 8 years old at WIL-31 entry
- Weight 23.5 kg
- wilate® prophylaxis for 12.3 months
  - 3× week during study
  - Mean weekly dose during the study: 113.7 IU/kg

Outcomes During WIL-31

- **7 bleeds, all minor**
  - 2 spontaneous, 2 traumatic, 3 allergic rhinitis
- 6 of 7 bleeds were treated each with a single infusion of wilate®
  - Mean wilate® dose: 47.4 IU/kg per infusion

Bleed sites during WIL-31

- **71% were nose bleeds**
  - Mean duration 3.1 h, a 3.4-fold decrease vs on-demand



- **All target joints resolved**

Octapharma, data on file.

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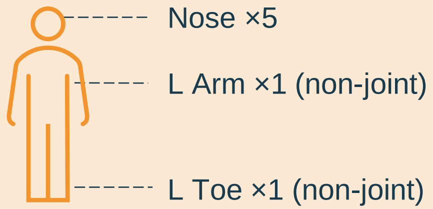
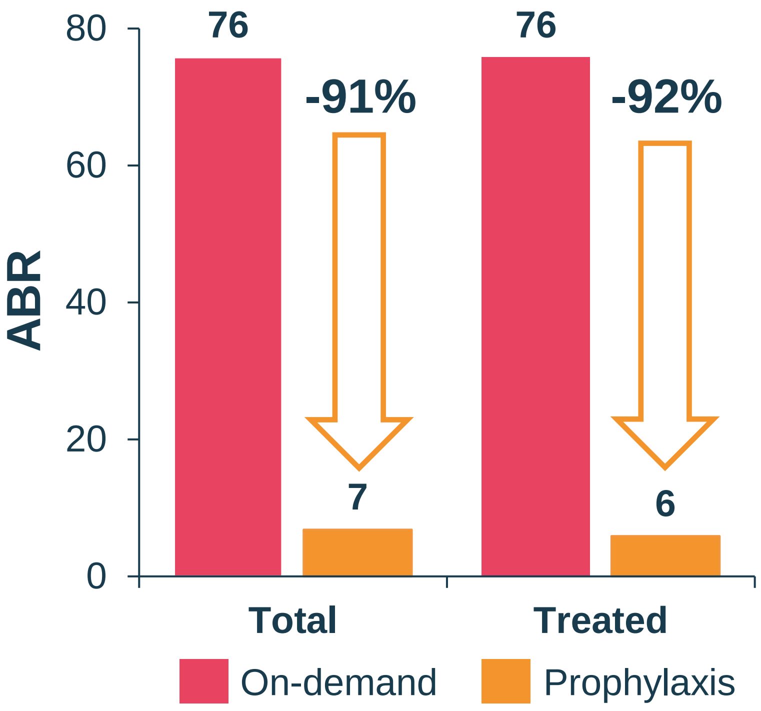


- **All target joints resolved**

wilate® prophylaxis reduced ABRs



Octapharma, data on file.

Ahmed's QoL* measures improved during WIL-29 and WIL-31

WIL-29 on-demand treatment (end of 6-month)

Anxiety: **often**
Focus: **sometimes**

Worst pain imaginable
(10/10)

Troubles with activities involving friends:
Always

I felt depressed**:**
sometimes

WIL-31 prophylaxis treatment (12-month visit)

Anxiety: **never**
Focus: **rarely**

Little pain
(1/10)

Troubles with activities involving friends:
never

I felt depressed**:**
rarely

* QoL measured by PROMIS-29 questionnaire.
Octapharma, data on file.

Summary

- **wilate® prophylaxis was efficacious in all children and adolescents** with all types of VWD
  - 89% and 85% reduction in total ABR for children and adolescents, respectively
  - 89% and 93% reduction in spontaneous ABRs for children and adolescents, respectively
- All (100% in children) or most (77% in adolescents) **bleeding events were minor during prophylaxis**
- **Children experienced no joint bleeds during prophylaxis**, and adolescents experienced reductions in knee and ankle joint ABRs compared to on-demand treatment
- Ahmed experienced a reduction in total and treated bleeds during prophylaxis and had improved QoL measures

UNDER THE SPOTLIGHT

Adult Living with Frequent Nosebleeds

Ana Boban

Haemophilia Centre, Department of Internal Medicine,
University Hospital Centre Zagreb,
Zagreb University School of Medicine, Zagreb, Croatia

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Marija has type 3 VWD and suffered frequent bleeds during on-demand treatment

Background / 6 months before the WIL-29 study

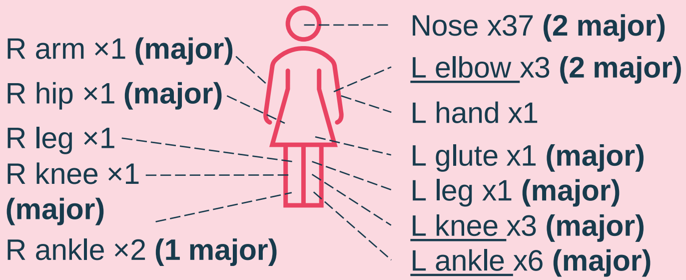
- Diagnosed with type 3 VWD in 1967 at the age of 2
- No family history of VWD
- 55-year-old female
- Weight 53 kg
- VWF activity (RCo): 0%
- Retrospective data: 24 bleeds treated on-demand with Immunate® over 6 months

Outcomes during WIL-29

- Total ABR of 115
  - 58 bleeds
  - Major bleeds (n = 19): 24 infusions, mean 21.3 IU/kg per infusion
  - Minor bleeds (n = 39): 40 infusions, mean 17.8 IU/kg per infusion
- 39 spontaneous bleeds, 19 traumatic
- 15 joint bleeds, 1 target joint (left ankle)

On-demand treatment during WIL-29

- Treated on-demand (with wilate®) for 6 months



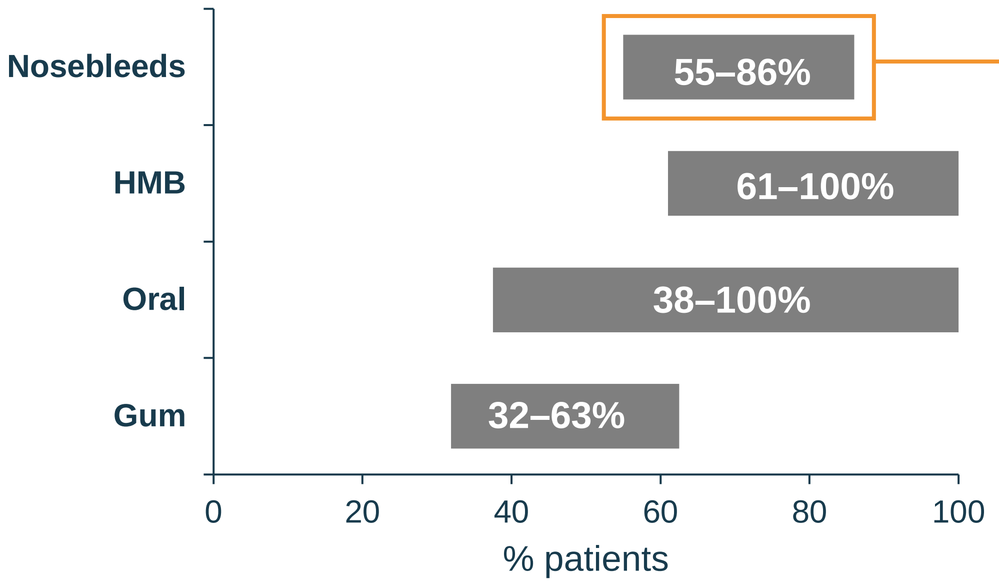
- 37 nosebleeds during 6-month period
  - **64% of bleeds**
  - **mean duration 3.3 h / nosebleed**
  - **5% of nosebleeds were major**

VWD: von Willebrand disease; IU: international units; L: left; R: right; RCo: ristocetin cofactor; VWF: von Willebrand factor; ABR: annualised bleed rate. Octapharma, data on file.

Nosebleeds are one of the most common bleed sites in patients with all types of VWD

Most frequently reported bleeding events[1]

Systematic review of 168 sources



Type 1: 39–53%

Type 2: 44–57%

Type 3: 37–77%

Nosebleeds and oral bleeds are the reason for initiating prophylaxis in

39%

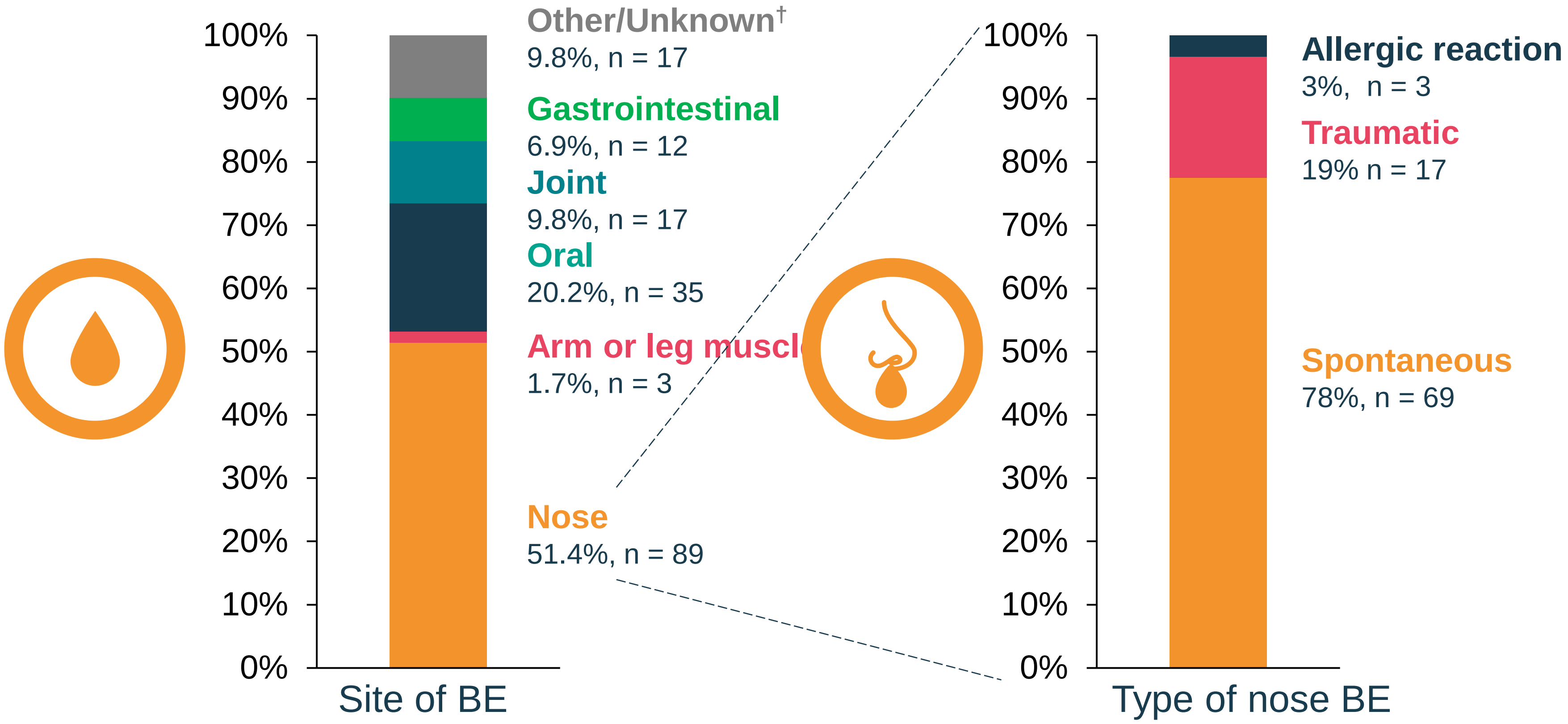
of adults and children (n = 130)[2]

"As soon as I took the packing out the nosebleed started again"[3]

"I got a really bad nosebleed, and I got, several that week, so I had to defer my exams"[4]

1. Du P et al. *J Blood Med* 2023; 14:189-208; 2. Rugeri L et al. *Eur J Haematol* 2022; 109:109-17; 3. Arya S et al. *J Thromb Haemost* 2021; 19:1506-14; 4. Arya S et al. *J Thromb Haemost* 2020; 18:3211-21.

WIL-31: Nosebleeds were the most common bleeding event

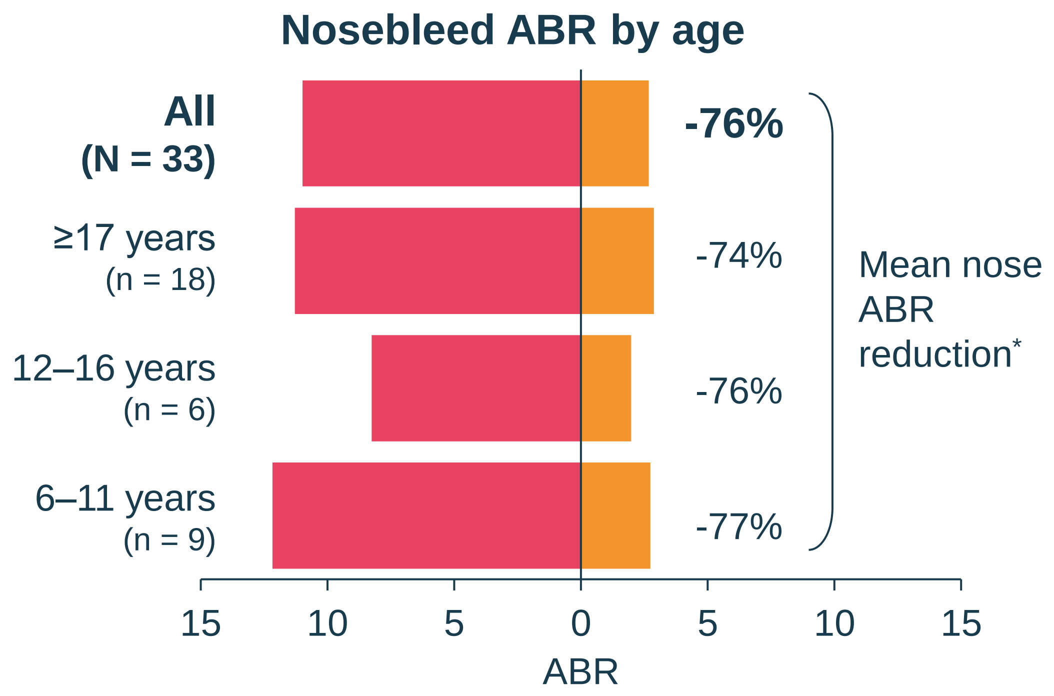


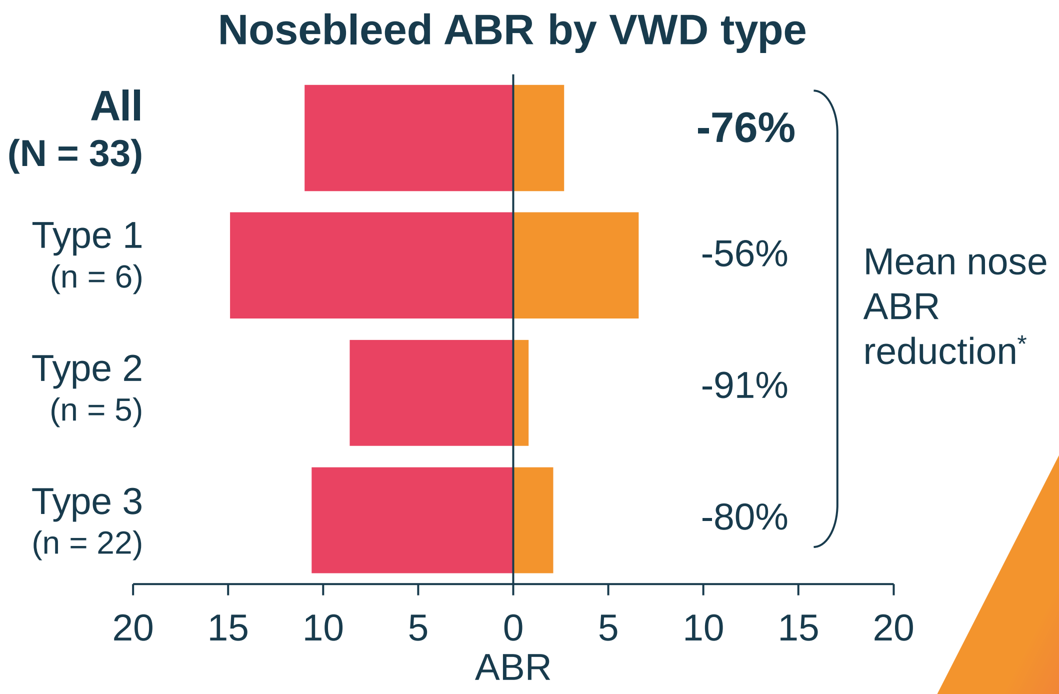
† Other bleeds in WIL-31 include cutaneous, haemorrhoid, head, hip, ocular, rectal, toe, and tonsil.
BE: bleeding event.
Octapharma, data on file.

WIL-31: wilate® prophylaxis substantially reduced nosebleeds versus on-demand treatment

Prophylaxis increased the percentage of patients with zero nosebleeds

On-demand: **21.2%** vs wilate® prophylaxis: **51.5%**





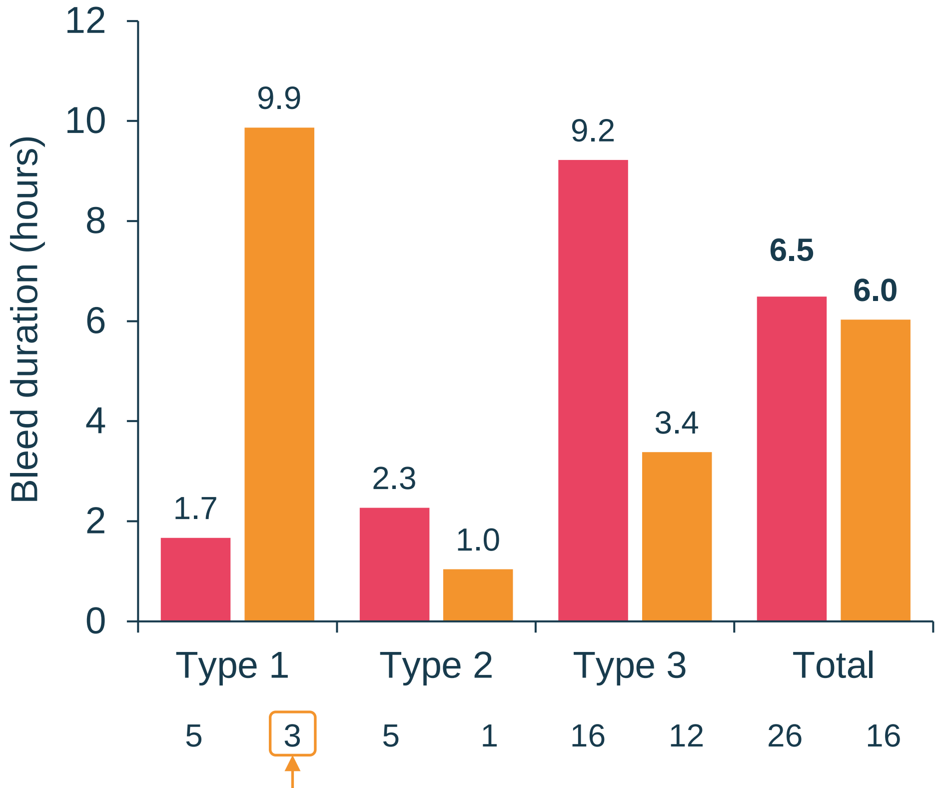
On-demand wilate® prophylaxis

*Reduction in mean ABR (%) = 100-(mean nose ABR WIL-31/mean nose ABR WIL-29*100). Missing in case nose ABR = 0 in WIL-29.
Octapharma, data on file.

WIL-31: wilate® prophylaxis reduces the duration of nosebleeds versus on-demand treatment

Mean duration of nosebleeds by age

Mean duration of nosebleeds by VWD type



On-demand wilate® prophylaxis

One patient, aged 52 years, experienced frequent nosebleeds (n = 20; 11 spontaneous, 9 traumatic) corresponding to a high mean duration (16.1 hours)

* Patients who experienced nosebleeds.
Octapharma, data on file.

WILPROPHY

WIL-31: wilate® was efficacious for the treatment of nosebleeds

Overall, 89 nosebleeds required treatment

Treated with wilate®	Required only a single infusion	Mean wilate® dose per event (IU/kg)
80.9% 72/89	**86.1%** 62/72	**47.2**

98.6% of treated nosebleeds were rated as "successfully treated"

Octapharma, data on file.

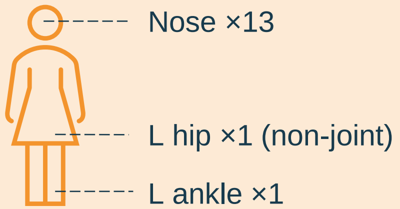
Marija experienced a reduction in rate and duration of bleeds during prophylaxis with wilate®

Prophylaxis with wilate® during WIL-31

- 55 years old at WIL-31 entry
- Weight 48 kg
- wilate® prophylaxis for 12 months
  - 2 infusions a week at study start, increased to 3 infusions a week
  - Mean weekly dose during the study: 88.4 IU/kg

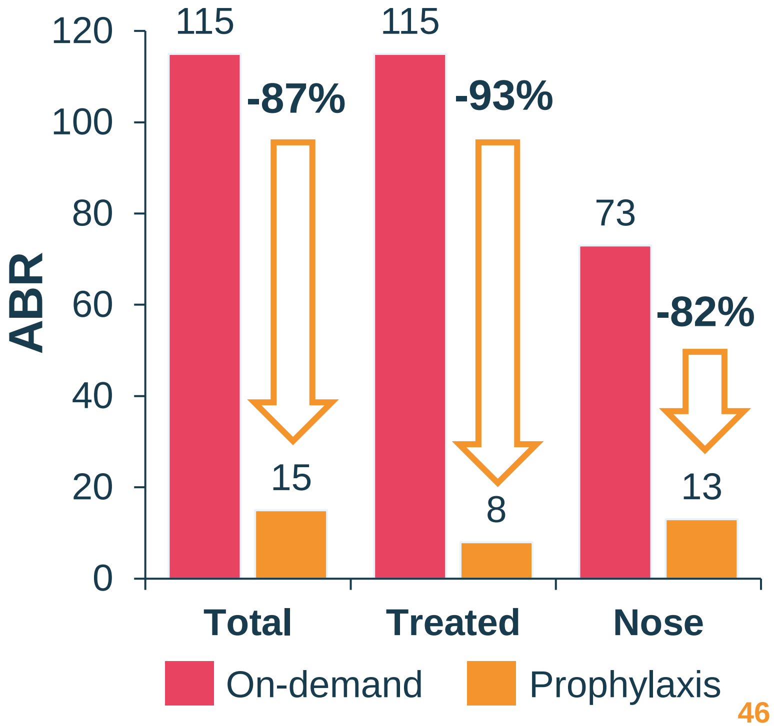
Outcomes during WIL-31

- 15 bleeds
- **All minor, all spontaneous**
- 87% were nosebleeds
  - **Mean duration 1.6 h / nosebleed, a 2-fold decrease vs on-demand**



- 13 nosebleeds during 1-year period

wilate® prophylaxis reduced incidence of BEs



Octapharma, data on file.

Summary

- **wilate® prophylaxis was efficacious in reducing nosebleeds** compared with on-demand treatment in all VWD types and age groups
  - Mean nose ABR reduced 76% during prophylaxis (11 vs 2.7 in WIL-29 and WIL-31, respectively)
  - More patients were free of nosebleeds during WIL-31 than WIL-29 (51.5% vs 21.2%)
- The **duration of nosebleeds decreased** for most age groups and VWD types
- 98.6% of nosebleeds during WIL-31 were **rated as successfully treated**
- Marija's experience:
  - a 82% reduction in nosebleeds
  - a 2-fold decrease in the duration of her nosebleeds during prophylaxis compared to on-demand treatment

UNDER THE SPOTLIGHT

Female Experiencing Heavy Menstrual Bleeding

Csongor Kiss

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Disclosures

Employment	
Research support	
Travel support	Novo Nordisk, SOB, CSL Behring
Consultancy	Novo Nordisk, Takeda, H-W-H
Major stockholder	
Speakers bureau	Novo Nordisk, Takeda, SOBI, Roche, CSL Behring
Honoraria	Novo Nordisk, Takeda, SOBI, Roche, CSL Behring
Scientific advisory board	
Patients	
Other	

Ilona has severe type 1 VWD and suffered from heavy menstrual bleeding (HMB) during on-demand treatment

UNDER THE SPOTLIGHT

Background / 6 months before the WIL-29 study

- 42-year-old female
- Severe type 1 diagnosed at 15 years old
- Family history of VWD
- Weight: 94 kg
- VWF: Activity (RCo): 16%; Antigen: 12%
- 7 bleeds treated on-demand*
- Birth control: levonorgestrel
- **Ongoing hypothyroidism, obesity, anaemia, type 2 diabetes, hypertension**

Outcomes during WIL-29

- Total ABR of 38†
  - 19 non-menstrual bleeds, all minor
    - 18 spontaneous, 1 traumatic
- **Heavy menstrual ABR of 7.9**
  - 5 menstrual bleeds
    - 4/5 (80%) were HMB
    - Mean duration 7.4 d per bleed

On-demand treatment during WIL-29

- Haemate P® consumption over 6 months
  - 1/4 (25%) HMB required treatment:
    - 2 infusions
    - Mean 26 IU/kg per infusion
  - 15% (3/20) minor bleeds were treated:
    - 3 infusions
    - Mean 26 IU/kg per infusion
    - 3 nose bleeds

*n = 1 menstrual (treated twice), n = 1 perioperative, n = 5 postoperative. †Excludes menstrual bleeding.
ABR: annualised bleeding rate; IU: international units; RCo: ristocetin cofactor; VWD: von Willebrand disease; VWF: von Willebrand factor.
Minor menstrual bleeds: menstruation of regular intensity. HMB (i.e. major): any menstrual bleeding that impedes the ability to perform daily activities such as work, housework; changing pads/tampons more frequently than hourly; menstrual bleeding lasting 7 or more days; and the presence of clots >1 cm combined with a history of flooding or a PBAC score ≥185.
Octapharma, data on file.

Heavy menstrual bleeding (HMB) and VWD

- HMB is excessive menstrual blood loss that interferes with a woman's physical, emotional, social and material QoL[1]
- HMB is the most common symptom observed in women with VWD[2]

In women with VWD, HMB is associated with significant morbidity[3]

HMB occurs in

~80%

of women with VWD[3]

QoL: quality of life; VWD: von Willebrand disease.

1. Heavy menstrual bleeding. Clinical guideline 88. London National Institute for Health and Clinical Excellence; May 2021;
2. Weyand AC et al. *Res Pract Thromb Haemost* 2021; 5:51-4; 3. Ragni MV et al. *Haemophilia* 2016; 22:397-402.

Burden of HMB in VWD

~9-fold more frequent hospitalisation than women with HMB without VWD[1]

Iron deficiency and anaemia iron deficiency[2]

11% of women with moderate/severe VWD required a **blood transfusion** at least once due to HMB[3]

Impact on health-related **quality of life**[4]

Interference with social activities, school and work[5]

1. Holm E et al. *Haemophilia* 2018; 24:628-33; 2. Munro MG et al. *Am J Obstet Gynecol* 2023; 229:1-9.; 3. de Wee EM et al. *Thromb Haemost* 2011; 106:885-92; 4. Rae C et al. *Haemophilia* 2013; 19:385-91; 5. Sanigorska A et al. *Res Pract Thromb Haemost* 2022; 6:e12652.

WIL-31: Impact of wilate® prophylaxis on HMB in women of childbearing age

N = 7

Females of childbearing age

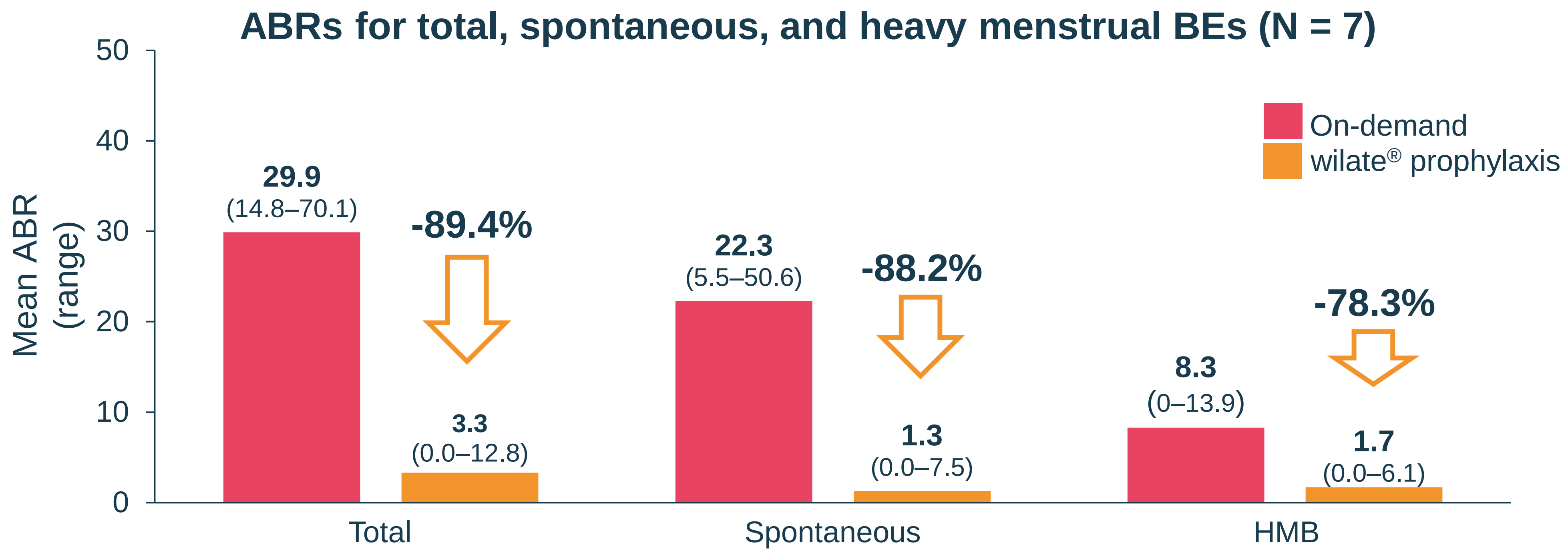
n = 6

Patients were taking hormonal birth control during WIL-31

Characteristic	N = 7
VWD type, n (%)	
Type 1	2 (29)
Type 2A	1 (14)
Type 3	4 (57)
Age, median, years (range)	18 (13–43)
Height, median, cm (range)	167 (163–170)
Weight, median, kg (range)	65 (53–94)
Race, n (%)	
Black or African American	1 (14)
Caucasian	6 (86)
Blood type, n (%)	
A	4 (57)
B	0 (0)
AB	0 (0)
O	3 (43)
Family history of VWD, n (%)	4 (57)

Octapharma, data on file.

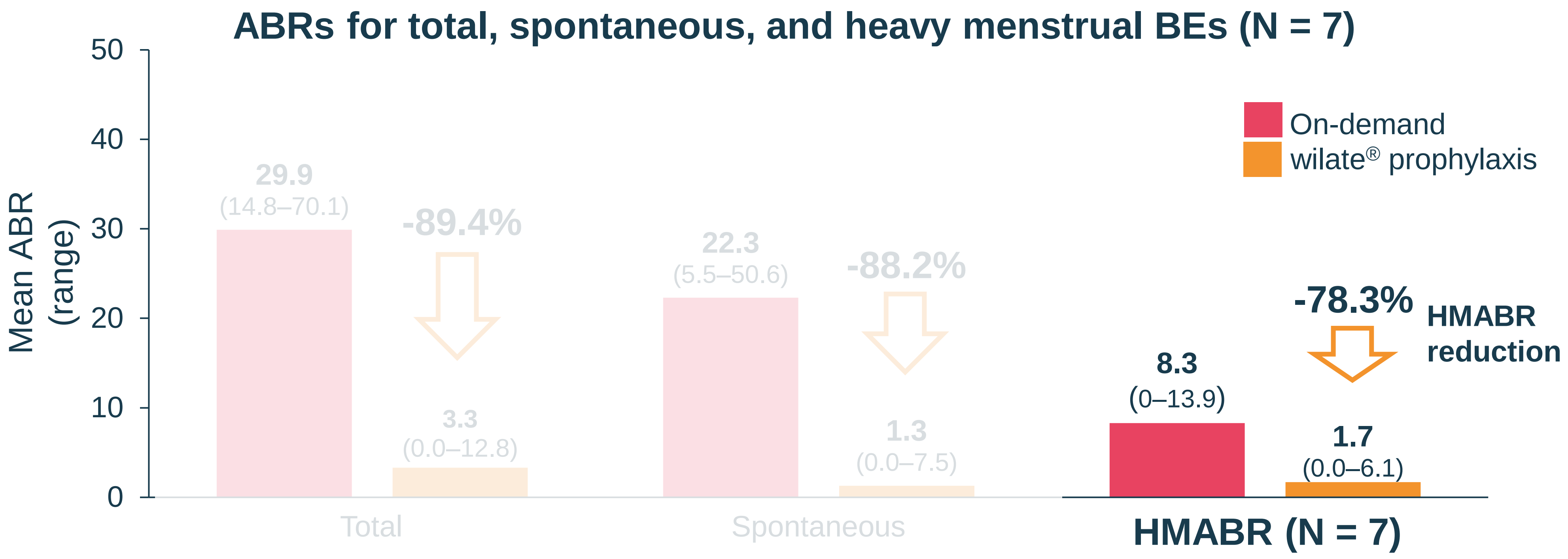
WIL-31: wilate® prophylaxis in women of childbearing age



Efficacy of wilate® was similar for females of childbearing age and the overall study population for reducing the occurrence of non-menstrual bleeding

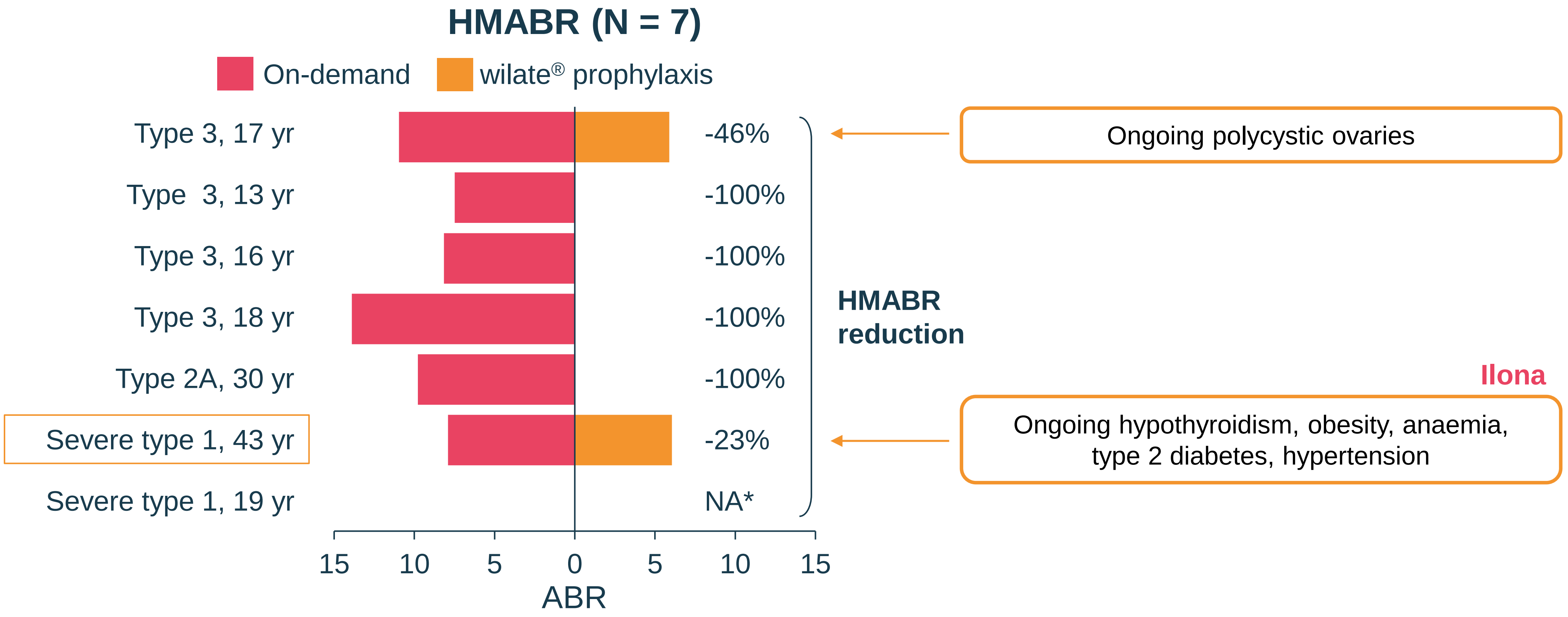
Octapharma, data on file.

WIL-31: wilate® prophylaxis in women of childbearing age



HMABR: heavy menstrual annualised bleed rate.
Octapharma, data on file.

WIL-31: HMB reduced in all women of childbearing age during wilate® prophylaxis



No additional treatment or hospitalisation was required for any HMB during prophylaxis

* No cases of HMB recorded in either study.
Octapharma, data on file.

HMB affects approximately one in four women[1] and is caused by a variety of disorders

International Federation of Gynecology and Obstetrics classification for the causes of abnormal uterine bleeding, including HMB[2,3]

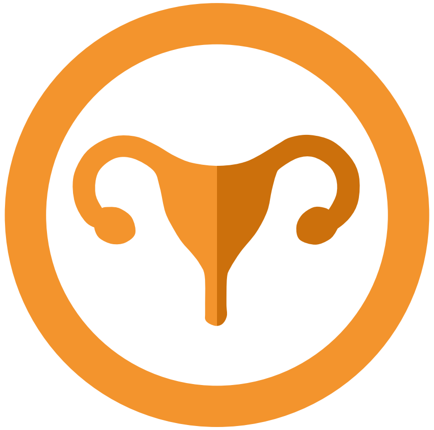
Structural causes

Polyps

Adenomyosis

Leiomyoma

Malignancy



PALM-COEIN

Non-structural causes

Coagulopathy

Ovulatory disorders
(e.g. PCOS, hyperthyroidism, obesity)

Endometrial

Iatrogenic

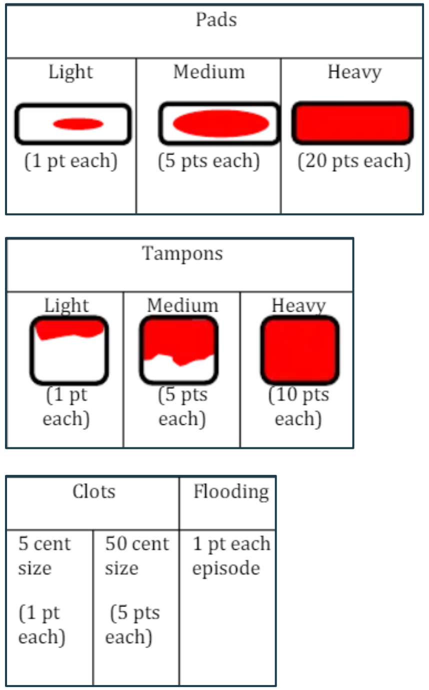
Not otherwise classified

PCOS: Polycystic ovary syndrome;

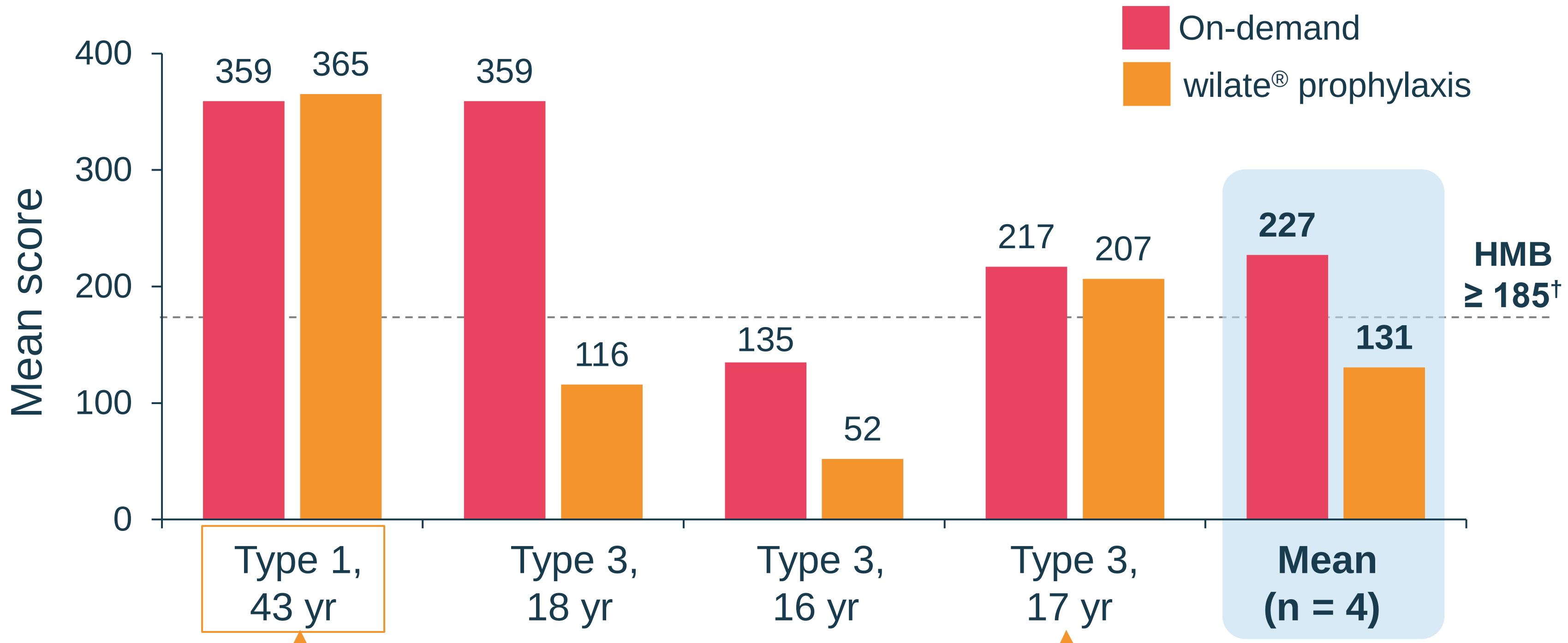
1. Pašalić E et al. *Pathol Res Pract* 2023; 251:154847; 2. Whitaker L and Critchley HOD. *Best Pract Res Clin Obstet Gynaecol* 2016; 34:54-65;
3. American College of Obstetricians and Gynecologists. Heavy Menstrual Bleeding (2022). https://www.acog.org/womens-health/faqs/heavy-menstrual-bleeding (accessed Jan 2024)

WIL-31: wilate® prophylaxis reduced HMB severity

PBAC and scoring system



Mean PBAC scores for individual patients*[1] (n = 4)



Ilona

Ongoing hypothyroidism, obesity, anaemia, type 2 diabetes, hypertension

Ongoing polycystic ovaries

*Only patients with PBAC for both WIL-29 and WIL-31 shown. †PBAC scores ≥185 correlates with HMB (>80 mL of menstrual blood loss).
PBAC: Pictorial blood assessment chart. Octapharma, data on file.

Ilona's menstrual bleeding episodes during wilate® prophylaxis

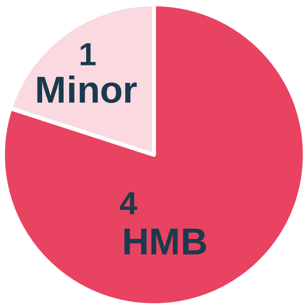
UNDER THE SPOTLIGHT

Outcomes During WIL-31

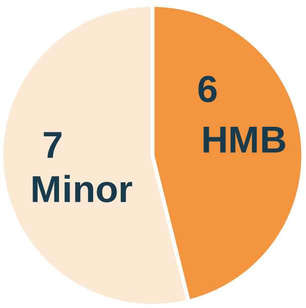
- **13 menstrual bleedings**, 6 HMB
  - Mean duration 7 d per bleed
  - Only 1 menstrual bleed impeded daily activities
- **No menstrual bleeds needed treatment**
- Birth control: levonorgestrel
- Mean PBAC score:
  - On-demand: 359
  - Prophylaxis: 365
- **Ongoing hypothyroidism, obesity, anaemia, type 2 diabetes, hypertension**

Severity of menstrual bleeds

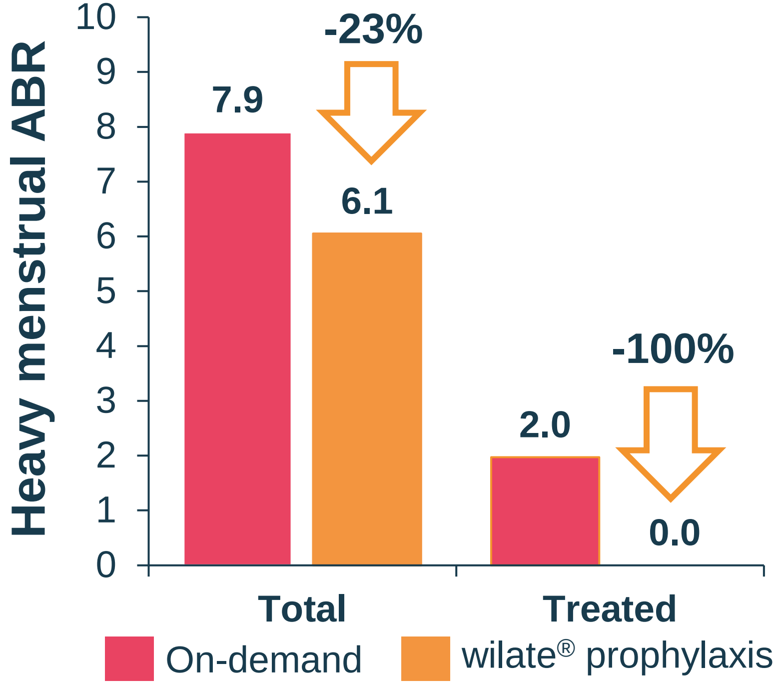
On-demand



Prophylaxis



wilate® prophylaxis reduced incidence of HMB despite co-morbidities



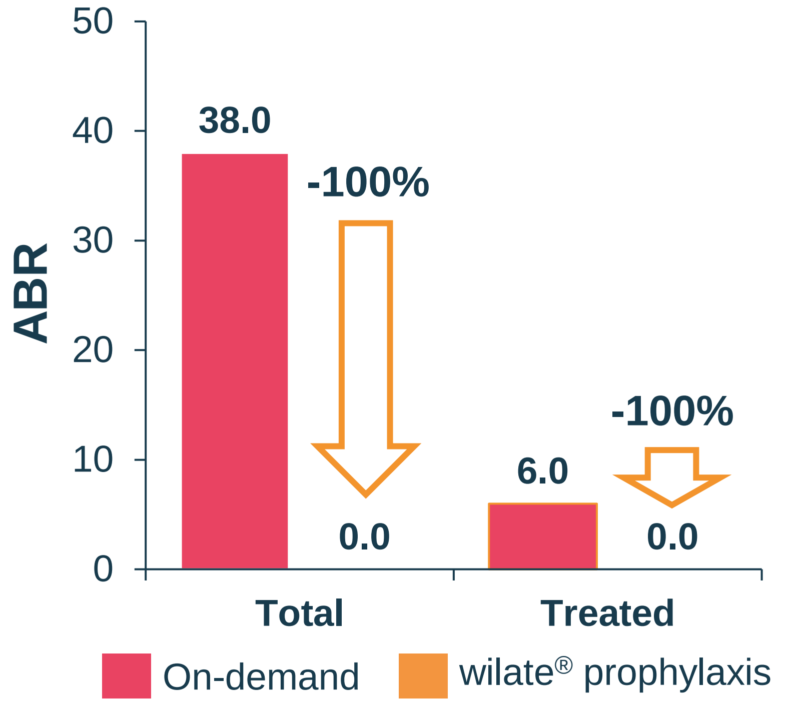
Octapharma, data on file.

Ilona experienced zero bleeds (non-menstrual) during prophylaxis with wilate®

wilate® prophylaxis resulted in zero non-menstrual bleeds

Prophylaxis with wilate® during WIL-31

- 43 years old at WIL-31 entry
- Weight 94.0 kg
- Haemoglobin: 125 g/L
- Platelets: 223 x10^9/L
- wilate® prophylaxis for 51.7 weeks
  - 2× week during study
  - Mean weekly dose: 48.5 IU/kg
- **Ongoing hypothyroidism, obesity, anaemia, type 2 diabetes, hypertension**



Octapharma, data on file.

Summary

- **wilate® prophylaxis during WIL-31 was efficacious in reducing HMB** compared with on-demand treatment in 7 patients with all types of VWD of childbearing age
  - Mean HMABR reduced 78% during prophylaxis (1.7 vs 8.3 in WIL-31 and WIL-29, respectively)
- Patients who still experienced some HMB while on prophylaxis had concomitant diseases which might affect the severity of menstrual bleeding
- During prophylaxis no episodes of HMB required additional treatment or hospitalisation
- Ilona experienced less severe menstrual bleeding during wilate® prophylaxis

UNDER THE SPOTLIGHT

Q&A

Please ask your questions now

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UNDER THE SPOTLIGHT

THANK YOU

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